

## **Analysing AMR in mobile genetic elements**

1. Using genomics to investigate antibiotic resistance and genomic features
2. Overview

As has been outlined in previous lectures and practical's genomics is an incredibly powerful tool to understand the molecular epidemiology, transmission and evolution of bacterial pathogens.

In the previous practical you investigate resistance using online tools that identify the presence of genes and resistance conferring mutations, we are now going to zoom down to individual bacterial genome which can help us further understand antibiotic resistance and extract data for further analysis.

Artemis is a genome browser and annotation software that was developed at the Wellcome Sanger Institute (<http://www.sanger.ac.uk/science/tools/artemis>). It is freely available to download for PCs and Mac.

The practical is designed to give you a basic understanding of Artemis software and to get a better idea of genome structure and content. It is not expected for you to master this software in a single session.

Important note – Artemis has a huge number of features for many different tasks – we are just concentrating on the basics – so don't worry about most of what is there.

If you have any questions during the practical, please ask!

## Analysis of resistance in genomes

### Comparative genomics

In this practical we are going to do a comparative genomics analysis two *Staphylococcus aureus* genomes, this to demonstrate the value of fine scale analysis of genomes and in particular mobile genetic elements (MGE) to understand AMR.

This analysis was published over 20 years ago in the early days of bacterial genomics but is a great example of how comparing two (or more) genomes can help understand the emergence and evolution of AMR. If you are interested in reading more here is the paper:

<https://www.pnas.org/doi/10.1073/pnas.0402521101>

The two isolates are:

1. MRSA252; this is a Methicillin-resistant *Staphylococcus aureus* (MRSA) strain that is **resistant** to Beta-lactam antibiotics.
2. MSSA476; this is a Methicillin-susceptible *Staphylococcus aureus* strain that is **susceptible** to Beta-lactam antibiotics.

Methicillin-resistance in *S. aureus* is primarily mediated by the acquisition of a gene called *mecA* which encodes an alternative penicillin binding protein 2a (PBP2a). The wildtype PBP2 is the target that Beta-lactam antibiotics bind stopping cell wall synthesis and repair by acquiring PBP2a – which is not bound by most Beta-lactam antibiotics MRSA strains can still grow in the presence of antibiotics.

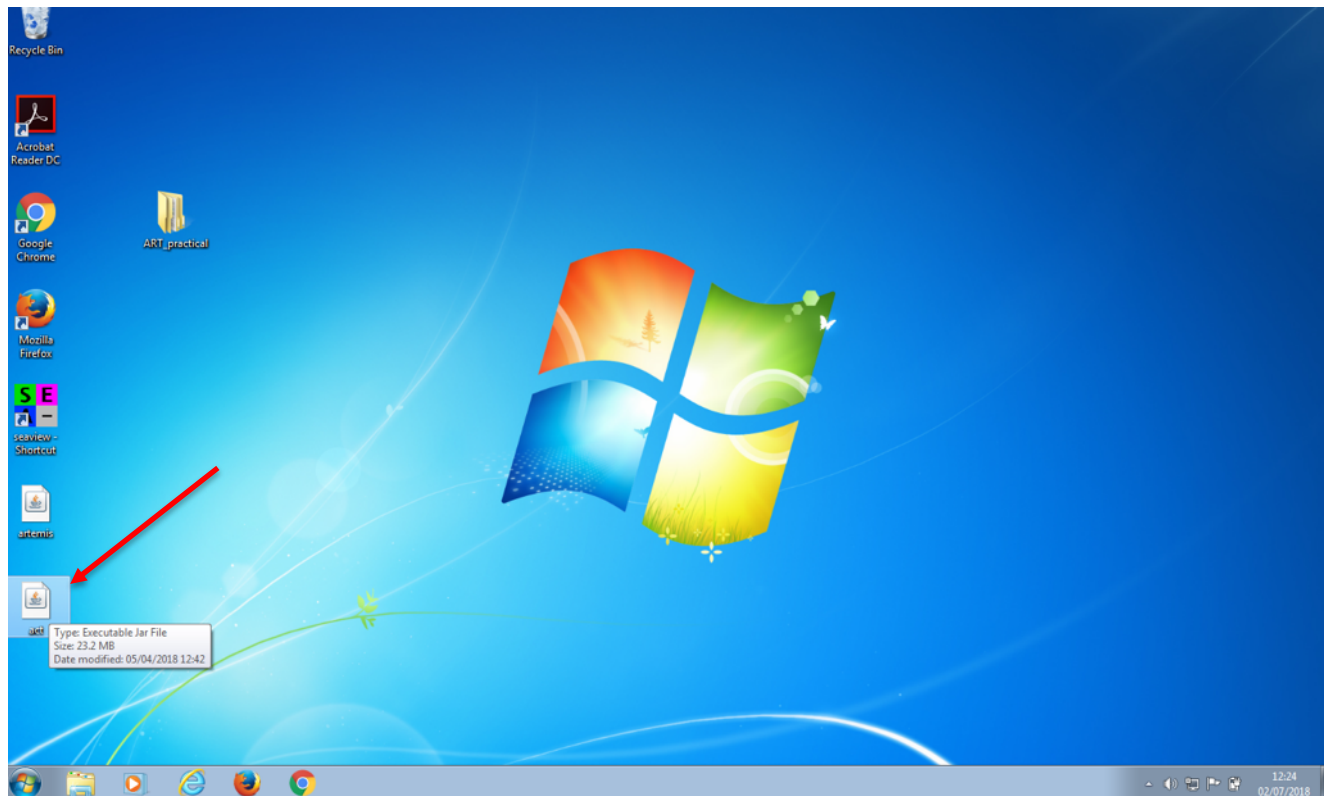
We are now going to investigate the *mecA* region in more detail to understand the origins and evolution of methicillin resistance in *S. aureus*. We are going to use a variation of the Artemis software called Artemis Comparison Tool (ACT):

Artemis Comparison Tool is a genome browser that lets you view comparisons of genomes. It was developed at the Wellcome Sanger Institute (<http://www.sanger.ac.uk/science/tools/artemis>). It is freely available to download for PCs and Mac.

The practical is designed to give you a basic understanding of the ACT software and to get a better idea of genome structure and content. It is not an expected for you to master using the software in a single session.

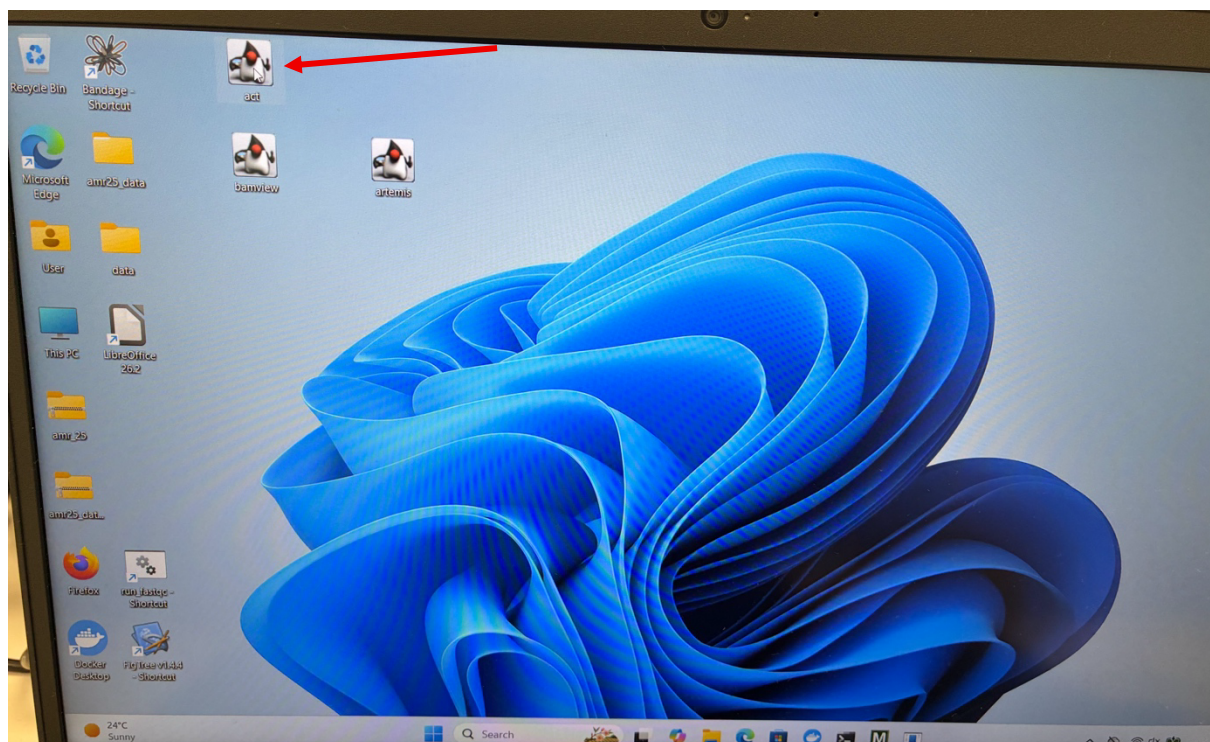
**Important note – ACT like Artemis has a huge number of features for many different tasks – we are just concentrating on the basics – so don't worry about most of what is there.**

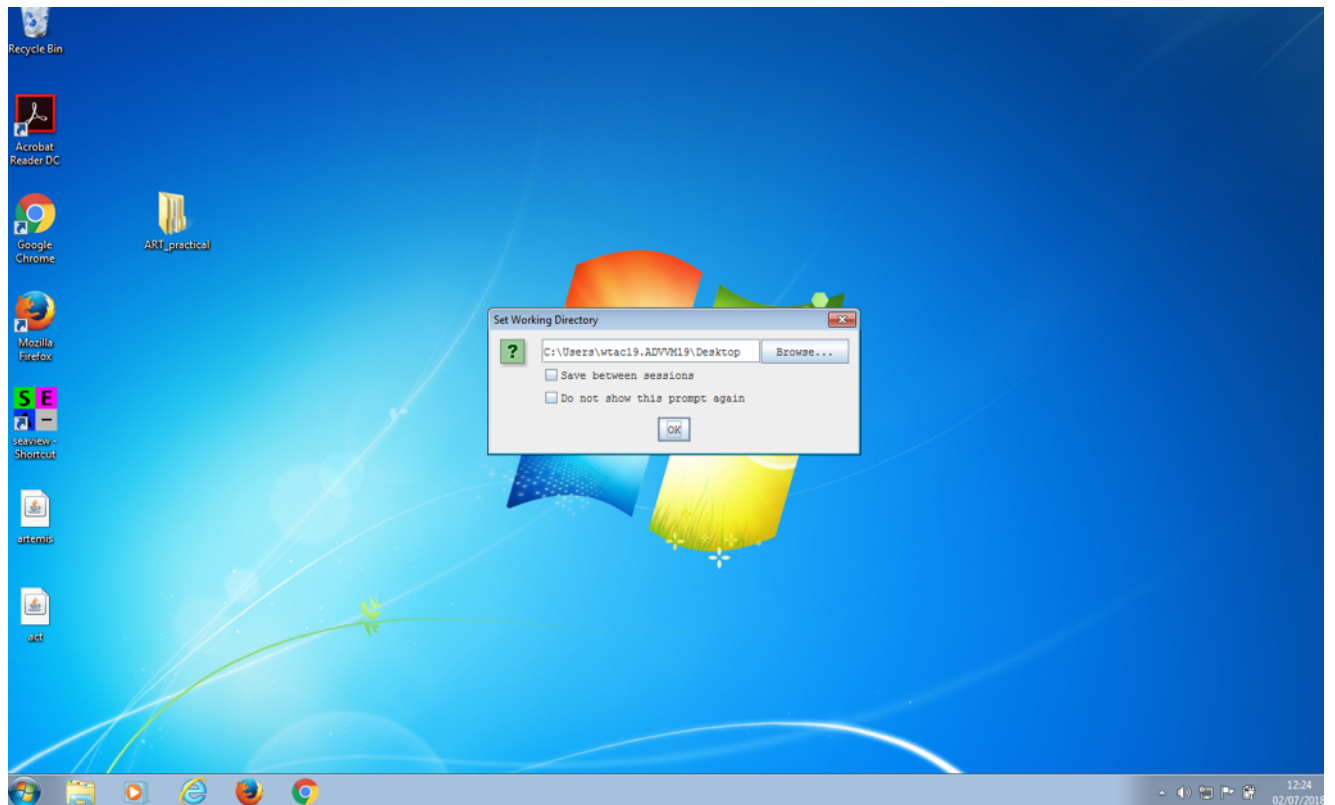
If you have any questions during the practical please ask!



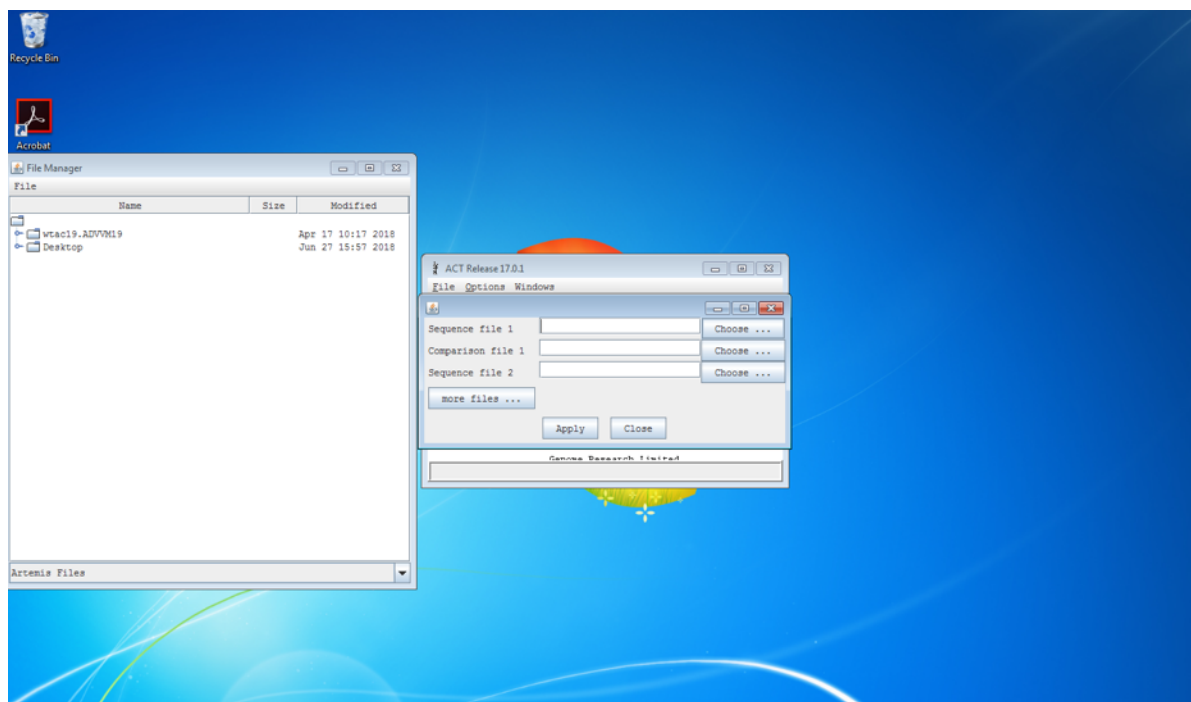
**Step 1:** Double click on the ACT symbol to open ACT. **NOTE:** if you cannot locate 'act' icon on your desktop, open up a new terminal window and type in 'act'. This will also open ACT tool too.

The ACT symbol on your desktop will look more the one below:



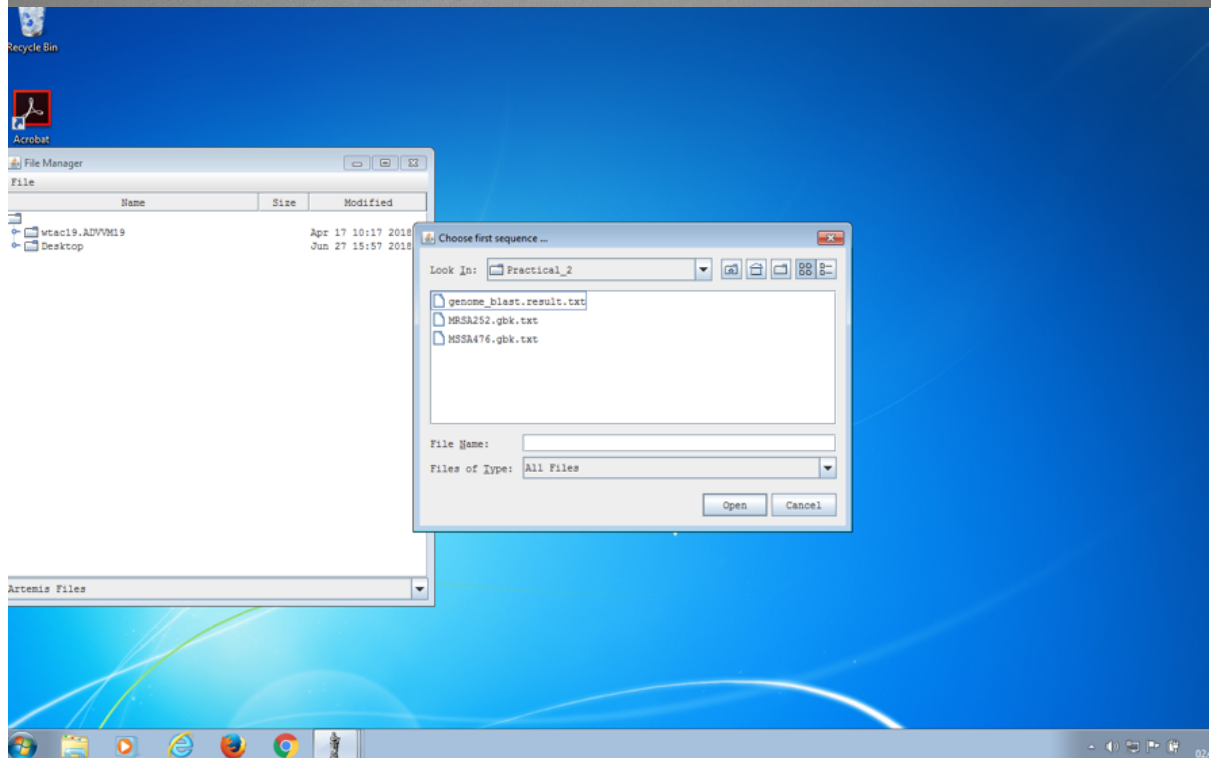
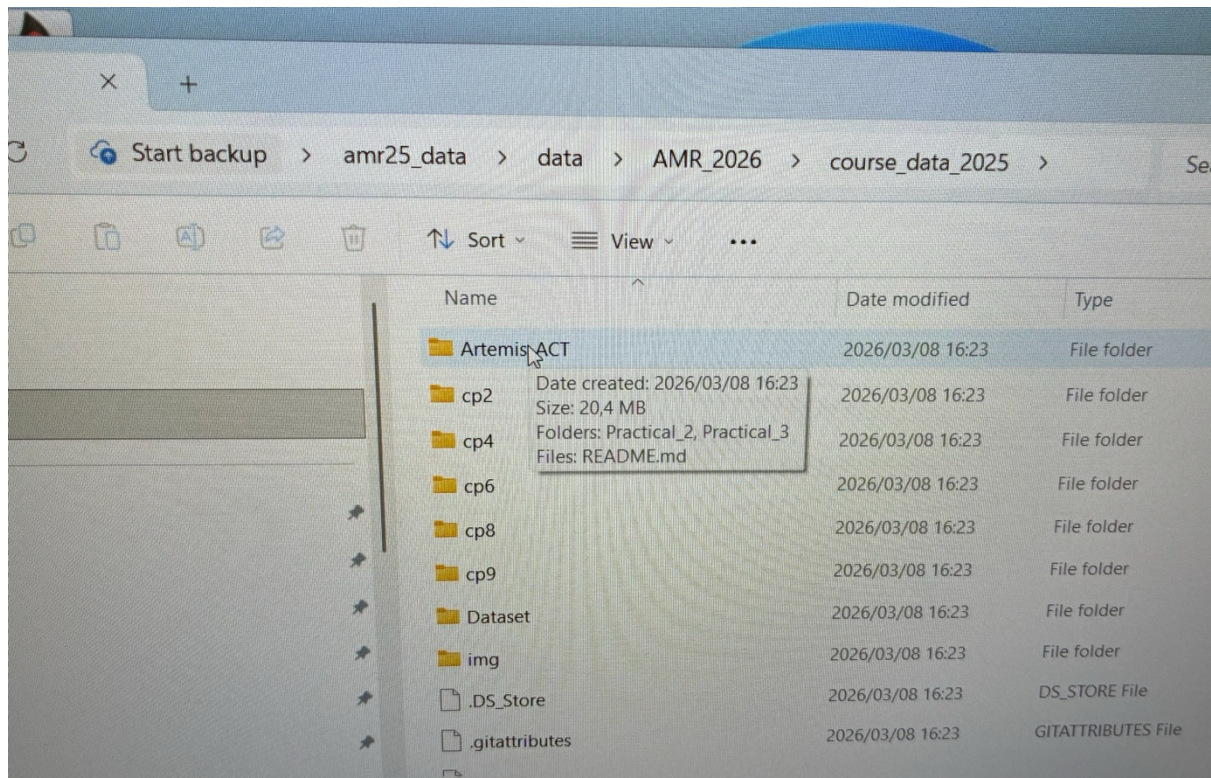


**Step 2:** A dialog box will appear. Click 'OK'.

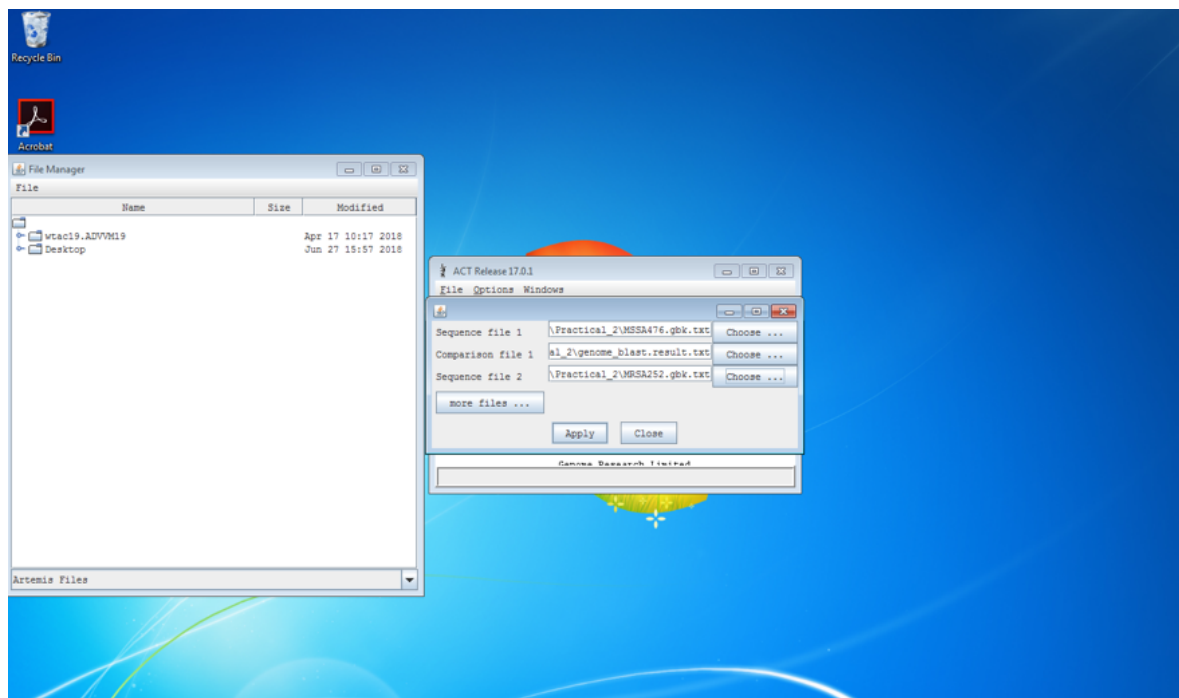


**Step 3:** We are now going to look at two genomes of *Staphylococcus aureus*; MSSA476 - a methicillin-sensitive strain and MRSA252 – methicillin-resistant strain. Click 'File' and then select 'Open ...'. A box will open like the one shown above. For Sequence file 1 – \_click the 'Choose ...' box. Navigate to the directory **amr25data/data/AMR2026/course\_data2025** then click the 'Artemis\_ACT' folder, and then open the Practical 2 folder. See next page for the folder on your screen.

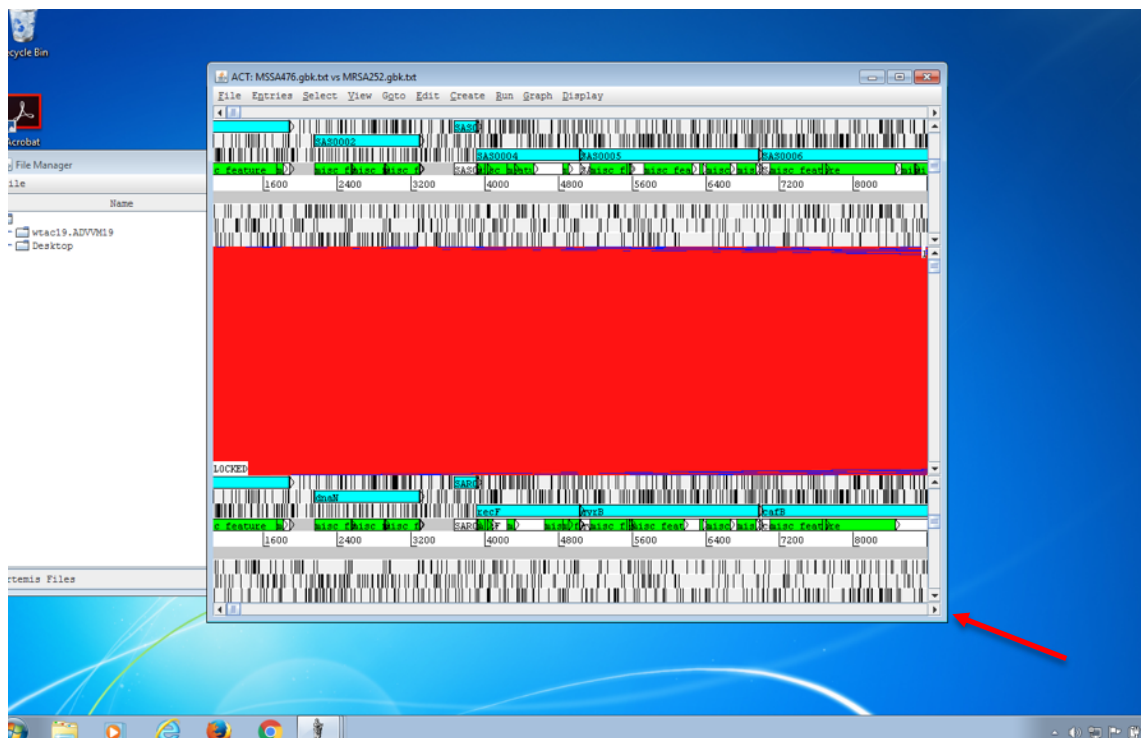




**Step 4:** The folder contains three files, these are the two genome sequences (MSSA476.gbk.txt, MRSA252.gbk.txt ) and a blast results file of the two genomes (MSSA476\_vs\_MRSA252\_blast\_results.txt). Click MSSA476.gbk.txt and click 'open'.



**Step 5:** Now click the 'Choose ...' button for 'Comparison file 1' and then select MSSA476\_vs\_MRSA252\_blast\_results.txt and click 'open'. Now do the same for Sequence file 1 selecting the file MRSA252.gbk.txt. Now press the 'Apply' button. A message will appear asking if you want view errors. Just click yes.



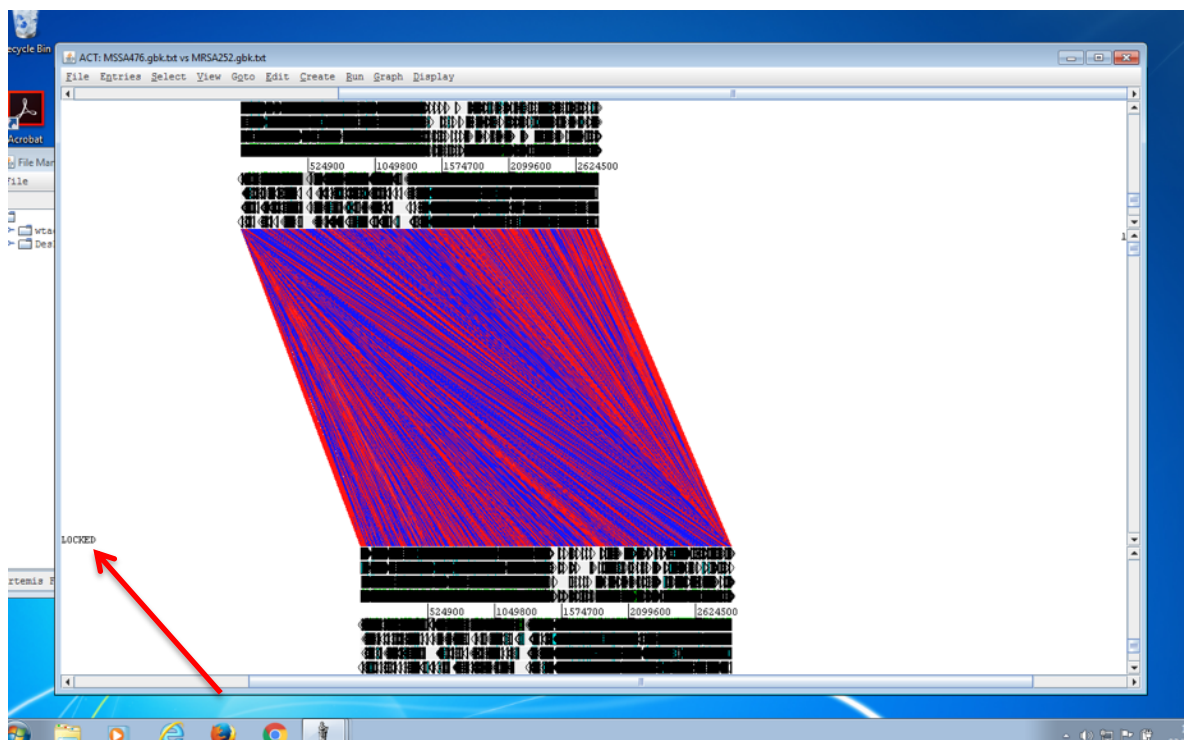
**Step 6:** A window like this will appear. Drag the corner to make the window bigger.



**Step 8: The ACT window is very similar to the Artemis window we used earlier but gives you a comparative view.**

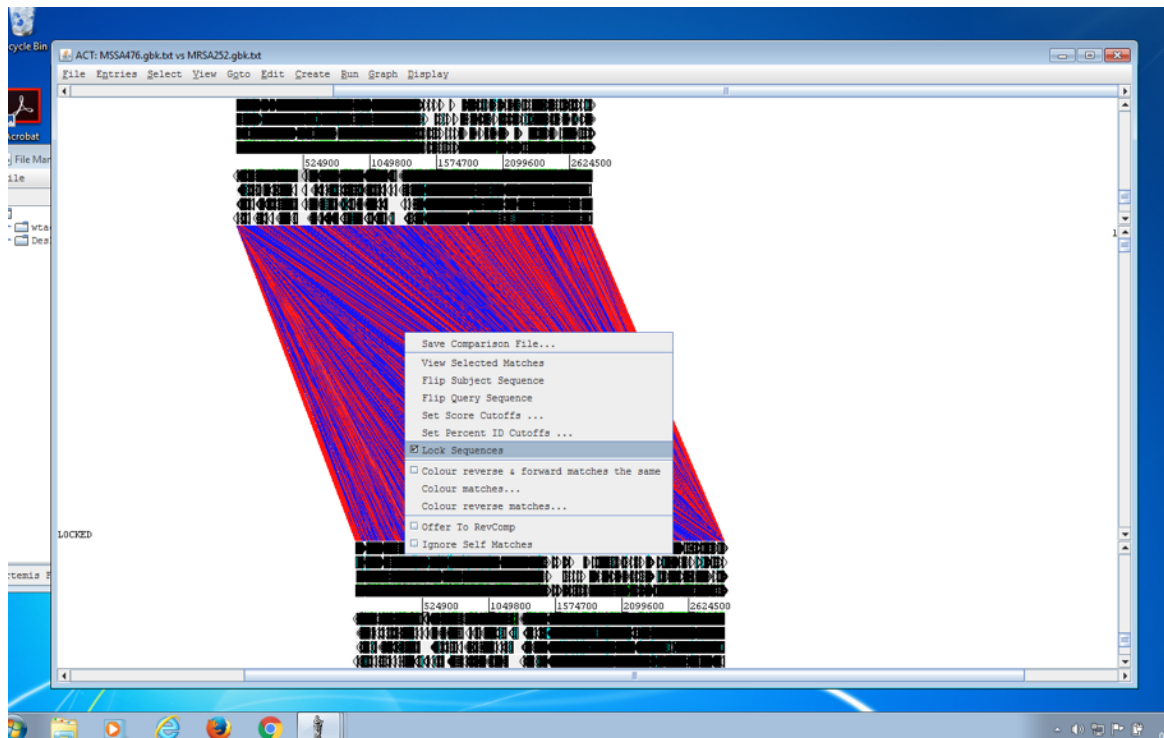
1. Drop-down menus: After clicking on a menu header you will then need to select a DNA sequence (in this case MSSA476 or MRSA252) before going to the full drop-down menu.
2. This is the Sequence view panel for 'Sequence file 1' (Subject Sequence – MSSA476) you selected earlier. It's a slightly compressed version of the Artemis main view panel. The panel retains the sliders for scrolling along the genome and for zooming in and out.
3. The Comparison View. This panel displays the regions of similarity between two sequences. Red blocks link similar regions of DNA with the intensity of red colour directly proportional to the level of similarity. Double clicking on a red block will centralise it. Blue blocks link regions that are inverted with respect to each other.
4. Artemis-style Sequence View panel for 'Sequence file 2' (Query Sequence – MRSA252).
5. Right button click in the Comparison View panel brings up this important ACT-specific menu which we will use later.
6. Sliders for zooming view panels.
7. Sliders for scrolling along the DNA.
8. Slider that allows you to filter the regions of similarity based on the length of sequence over which the similarity occurs.

**Step 9:** Use one of the zooming sliders (marked 6) and zoom the view all the way out to the maximum.

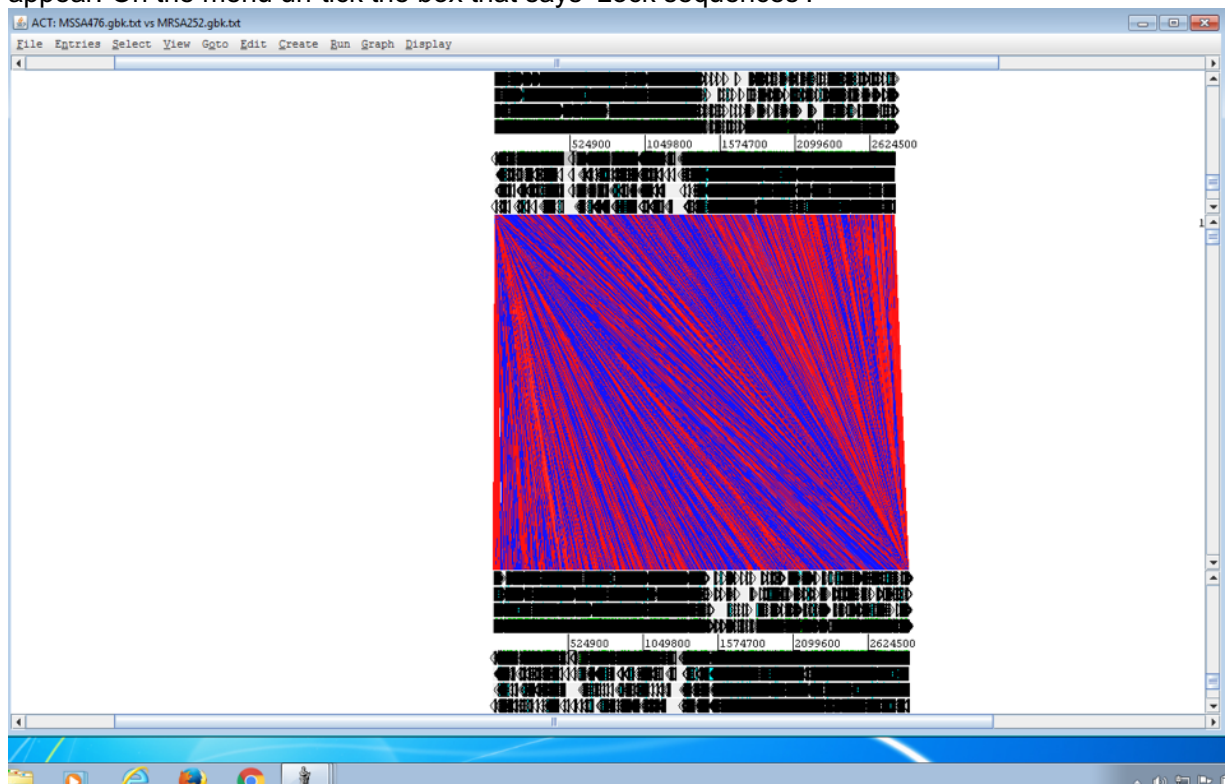


**Step 10:** You will notice here that it says 'LOCKED'. If you use one of the sliders to move down the genome – both genomes will move together. We are going to turn this off so the genomes move independently.

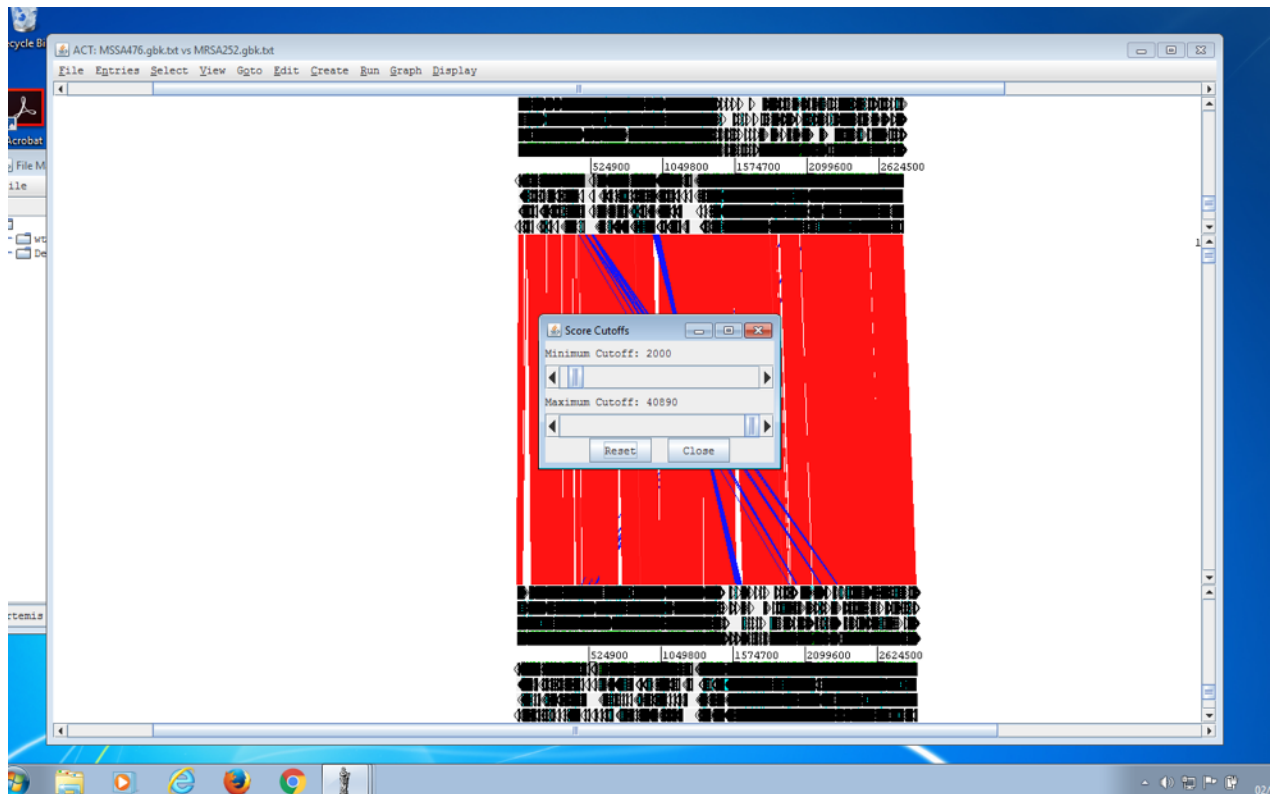




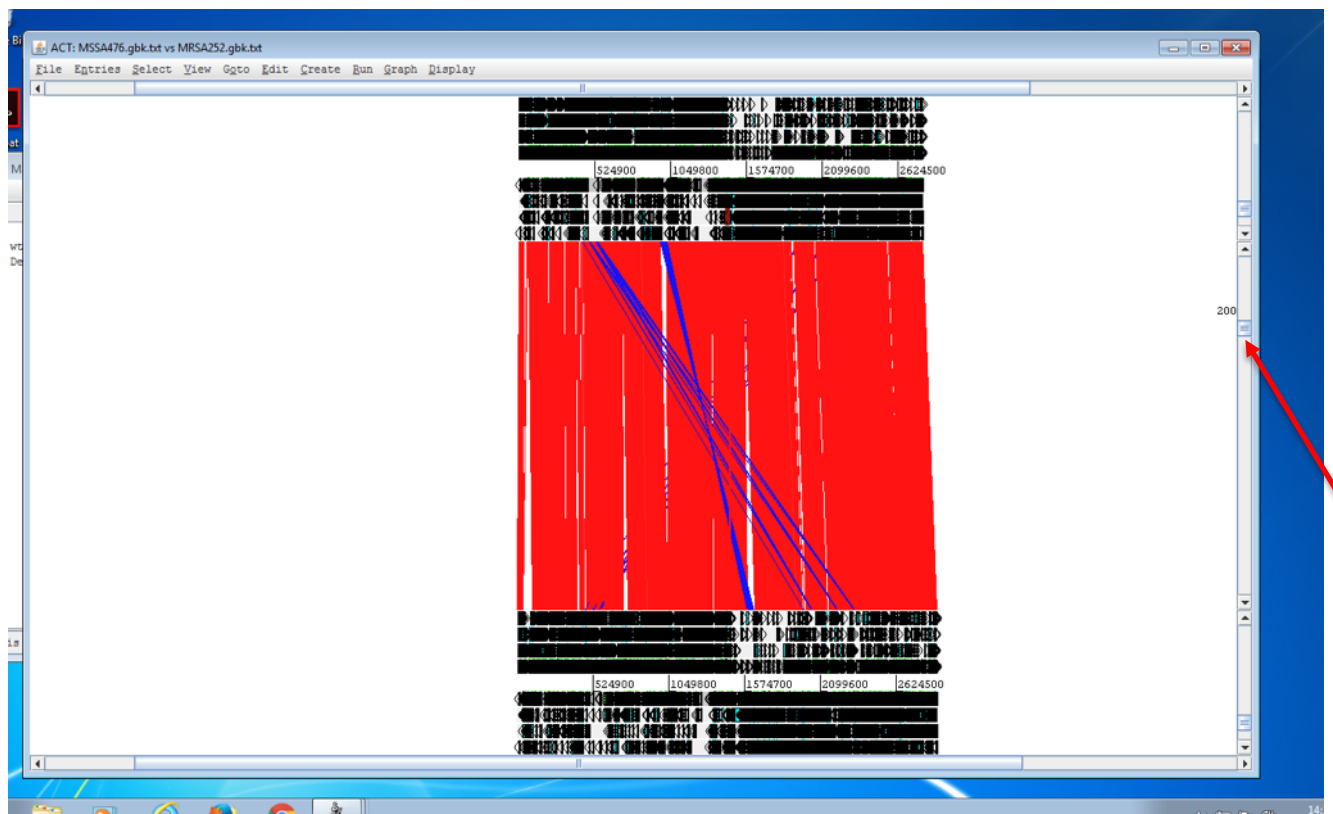
**Step 11:** Right click the mouse anywhere in the comparison area (the red / blue). A menu should appear. On the menu un-tick the box that says 'Lock sequences'.



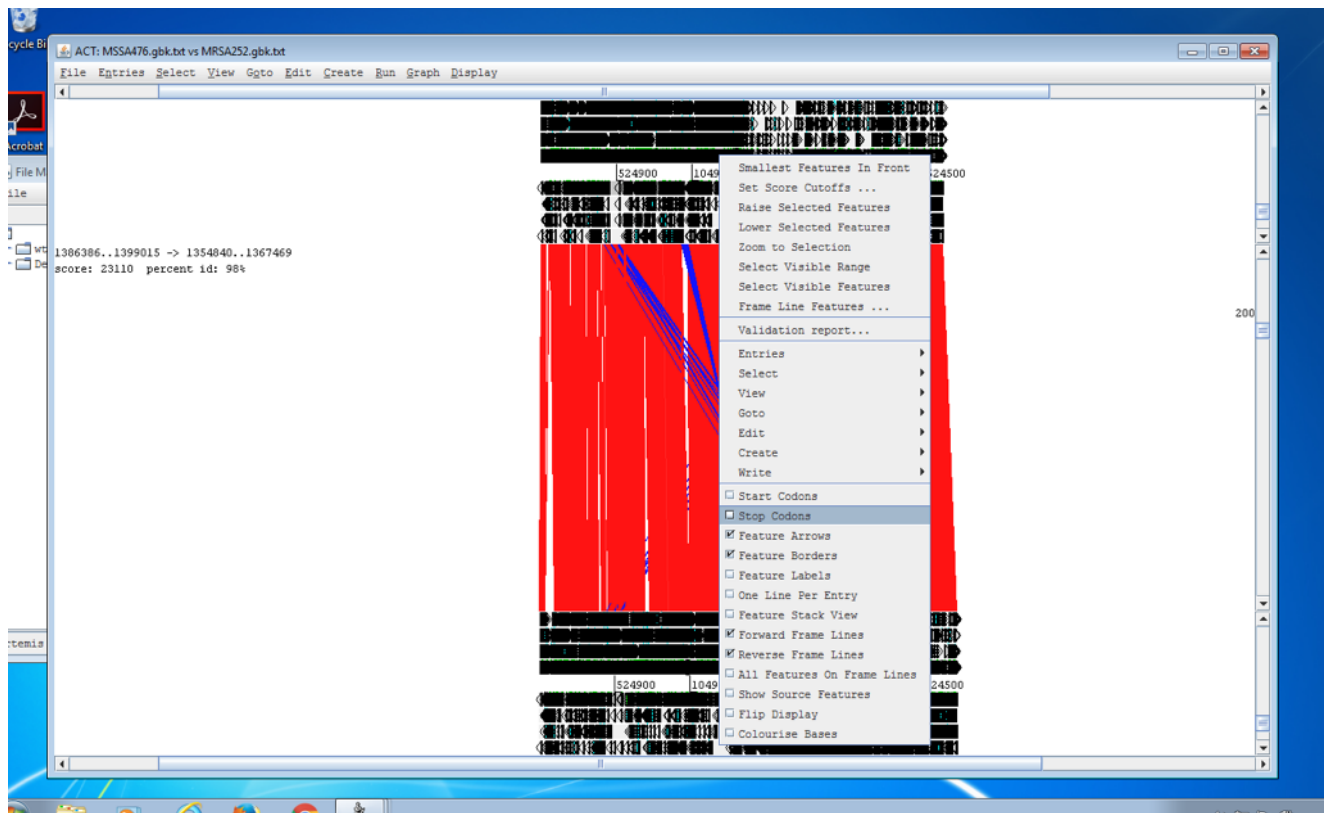
**Step 12:** The sequences will now move independently. Try moving them around yourself. Once you have tried, line them up again so it looks like the picture above.



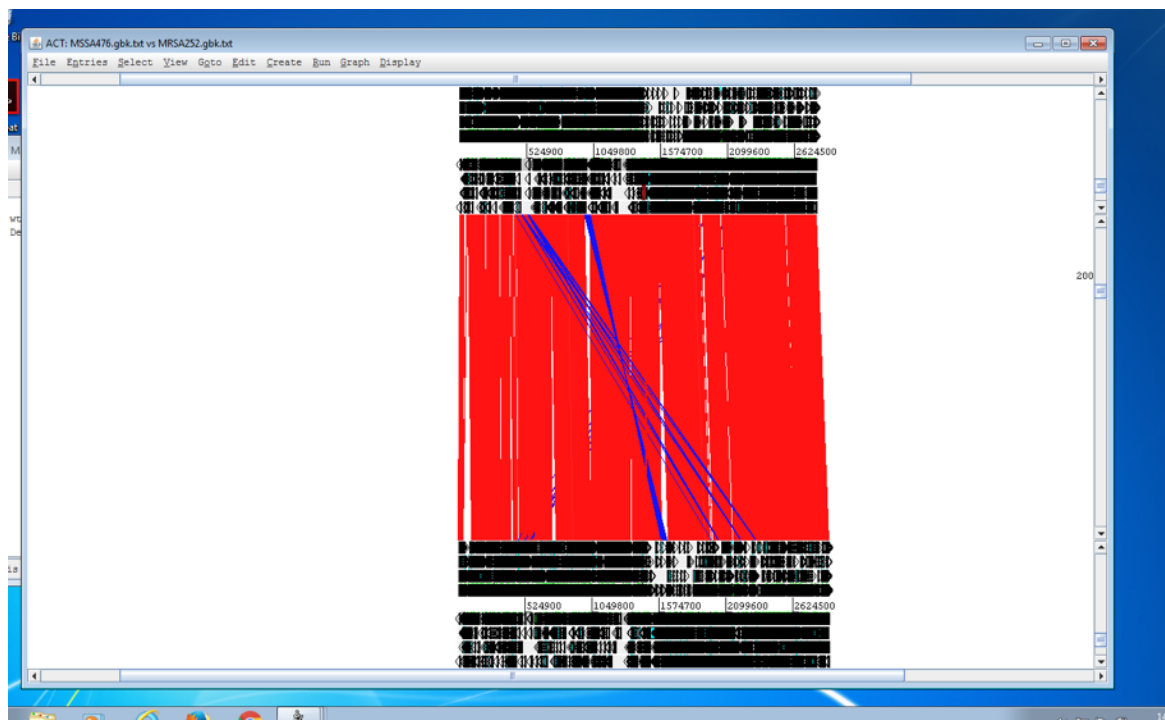
**Step 13:** Now, right click the mouse anywhere in the comparison area (the red / blue). Select the 'Set score cutoff'. Move the upper slider in the Score Cutoff of menu to 2000. This will remove any BLAST hits from the view with a score of less than 2000. Try moving the cutoff up and down to see what happens. Then leave it set at 2000 and click anywhere outside in the white to hide the menu or just move it out of the main window.



**Step 14:** Now slide the length of the sequence slider to 200. The window will now only show BLASTN hits longer than 200bp.

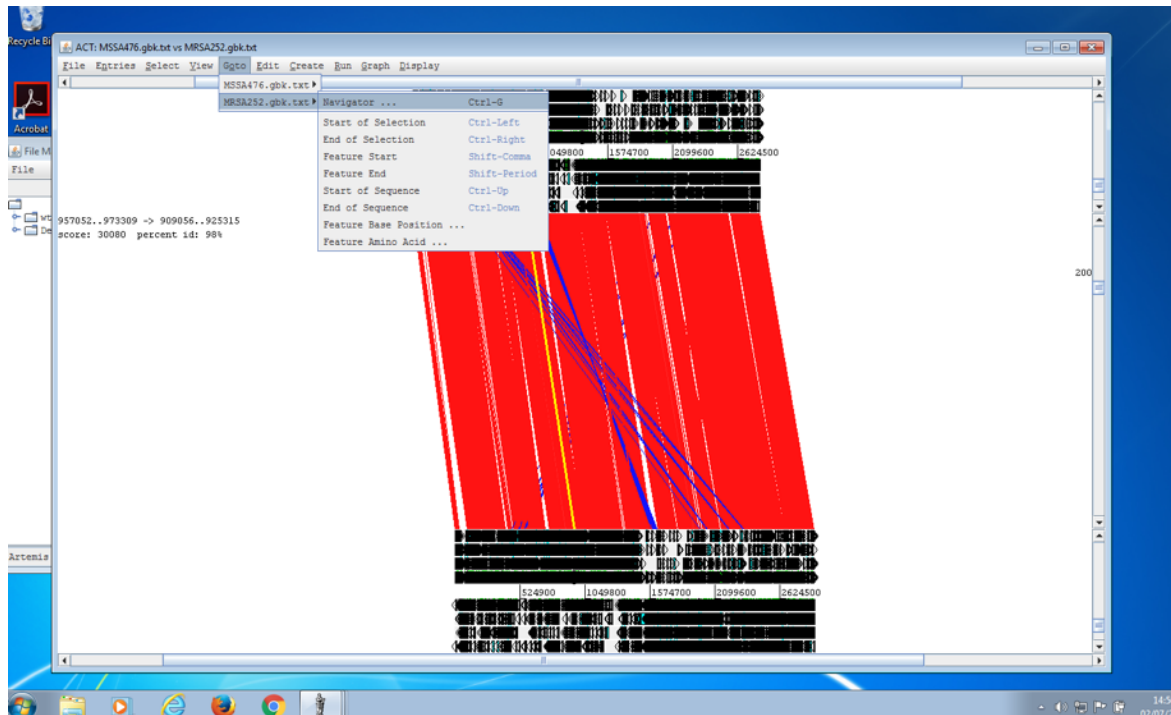


**Step 15:** Next, right click on the sequence viewer for the top sequence (MSSA476) and un-tick the 'Stop codons' box. Now do the same for the bottom sequence (MRSA252). We are now ready to explore the sequences.

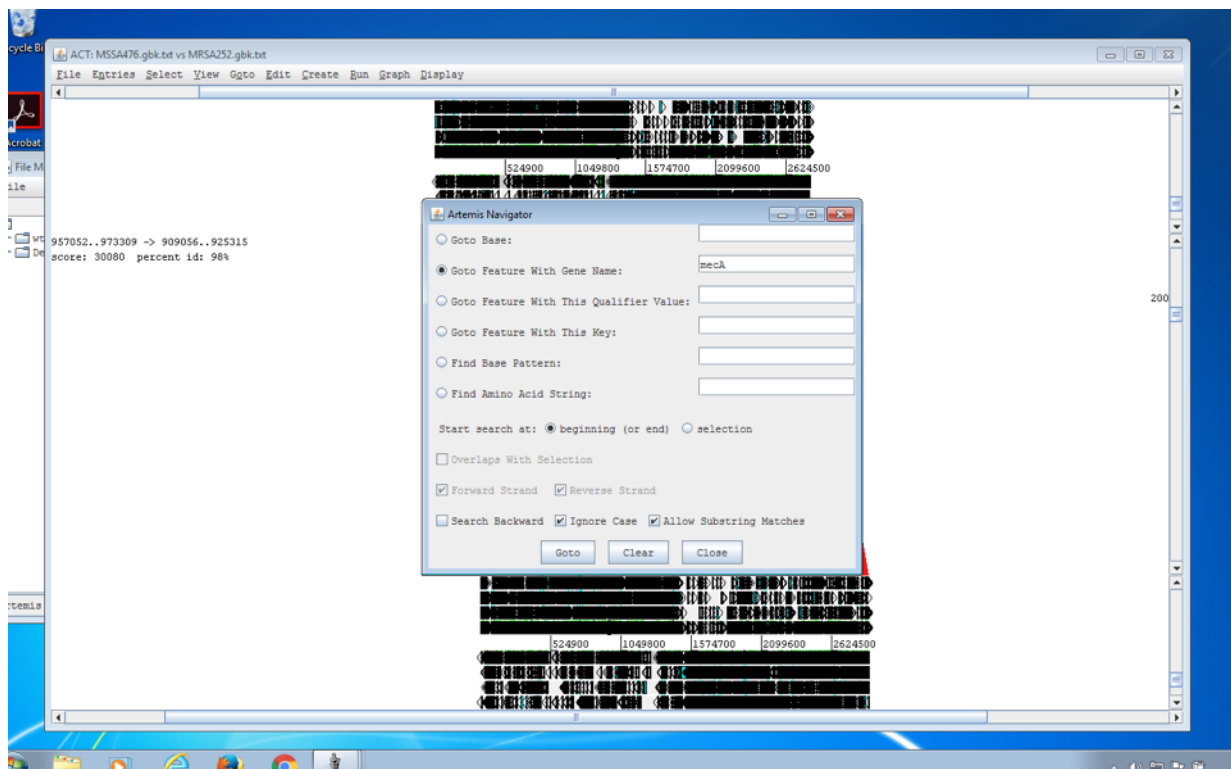


**Step 16:** We can see that the two genomes share a great deal of conservation and synteny (genes in the same order). You can also see a number of regions of difference between the two areas that are white in one genome but present in the other. If you use the slide bar we can zoom in to take

a closer look at these differences. Try this and take a look around the genome at some of the differences.

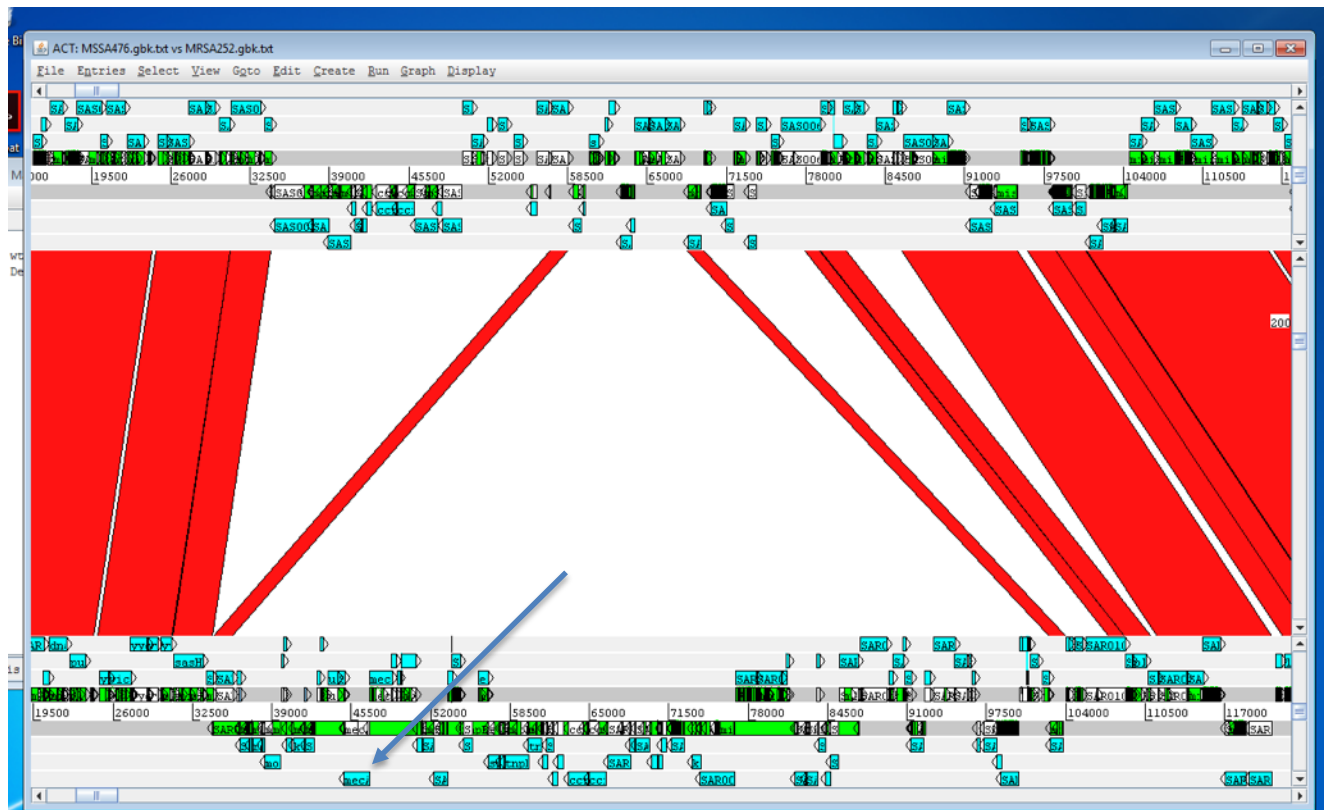


**Step 17:** We are now going to take a look at some of these differences that contain antibiotic resistance genes. Click on the 'Goto' menu at the top and select the lower sequence 'MRSA252.gbk.txt' and then click 'Navigator'.

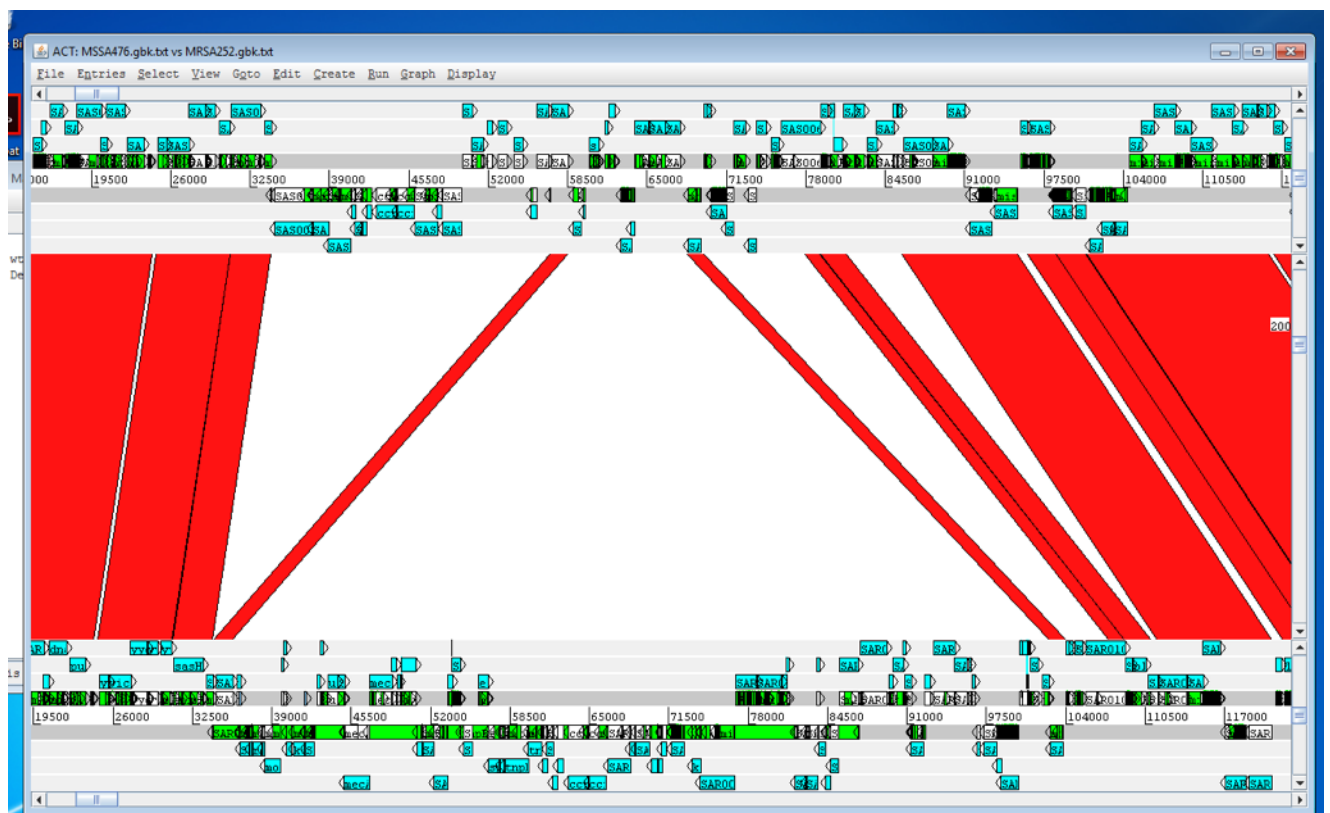


**Step 18:** Click the 'Goto Feature With Gene Name' and type 'mecA' in the box and then click 'Goto'.

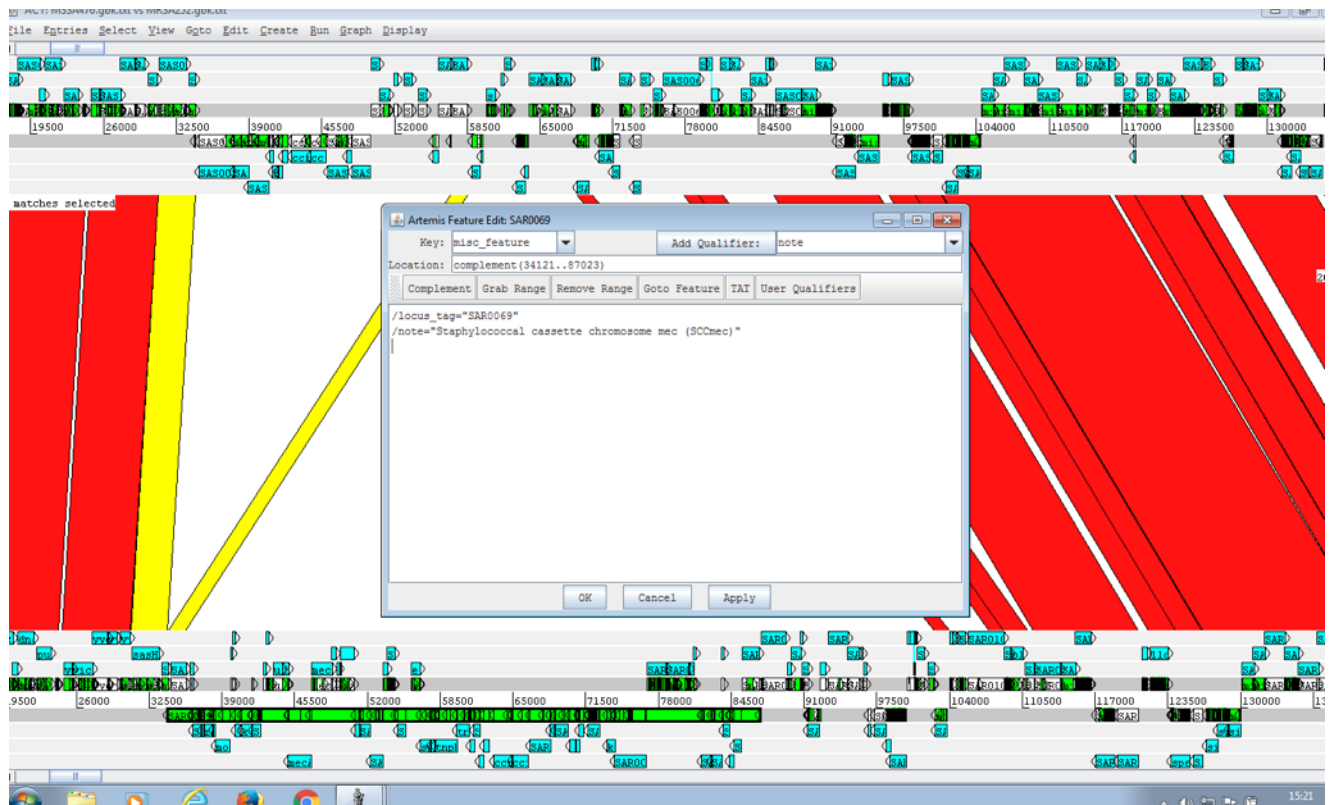




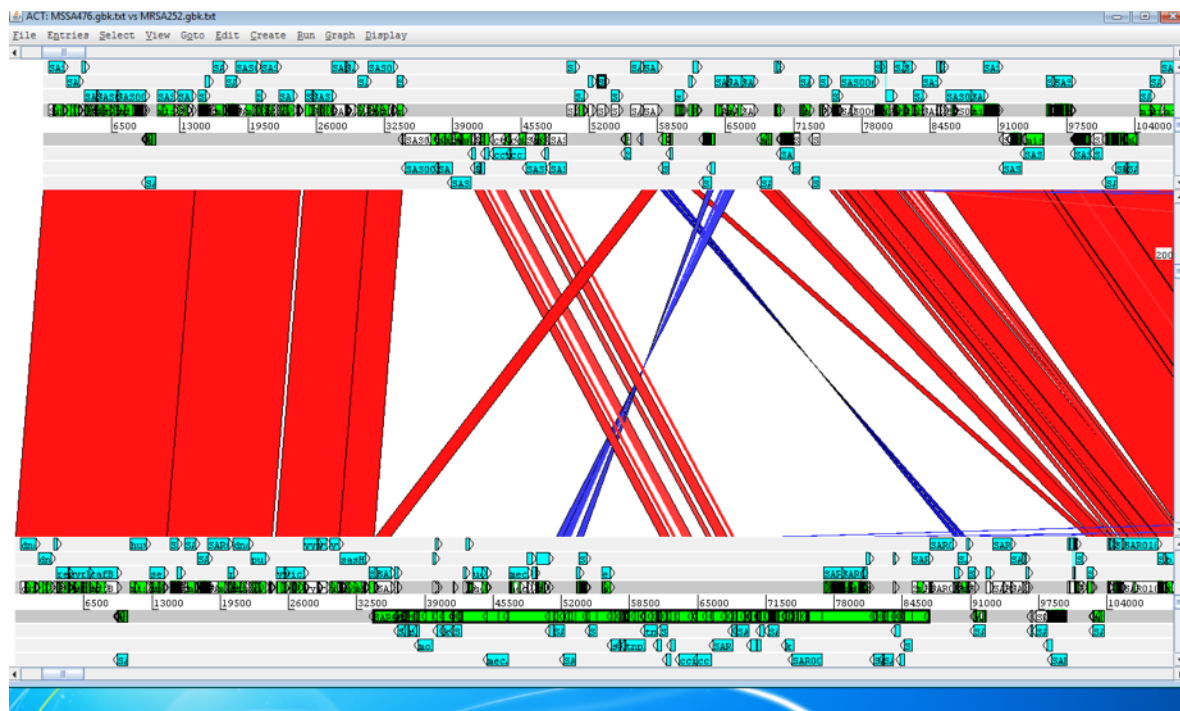
**Step 19:** Using the scrolling bars and zoom move the viewer until the window looks something like this. *mecA* should be highlighted (shown here with the arrows).



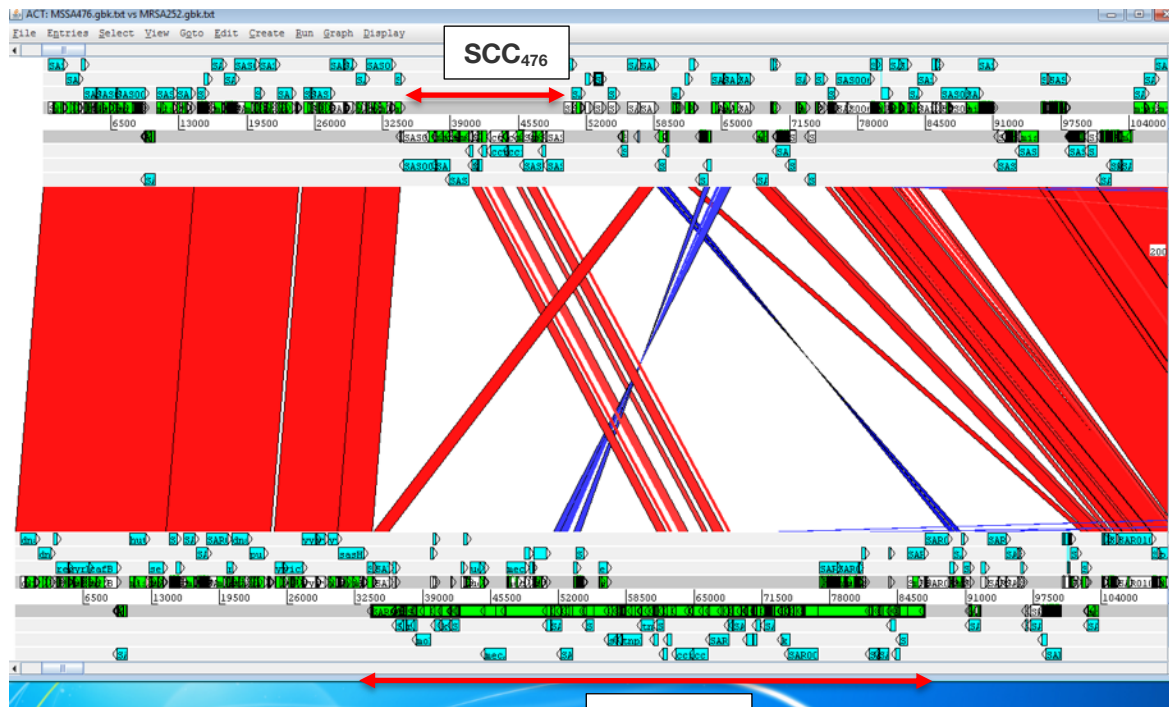
**Step 20:** If you click on the green region in the bottom genome, and then right click and select 'View' and 'Selected feature'.



**Step 21:** A menu will appear showing you that this region is a 'Staphylococcal cassette chromosome *mec*' also known as a *SCCmec*. This is a mobile element that brought the *mecA* gene into the strain.



**Step 22:** If right click the mouse anywhere in the comparison area (the red / blue). Select the 'Set score cutoff'. Move the slider down in the Score Cutoff of menu to back to 0. You should then see the above view. You can now see that in both genomes something has inserted in the chromosome and is flanked either side by conserved regions of the genome. This is why a comparative view is useful – it allows you to see the differences in context.



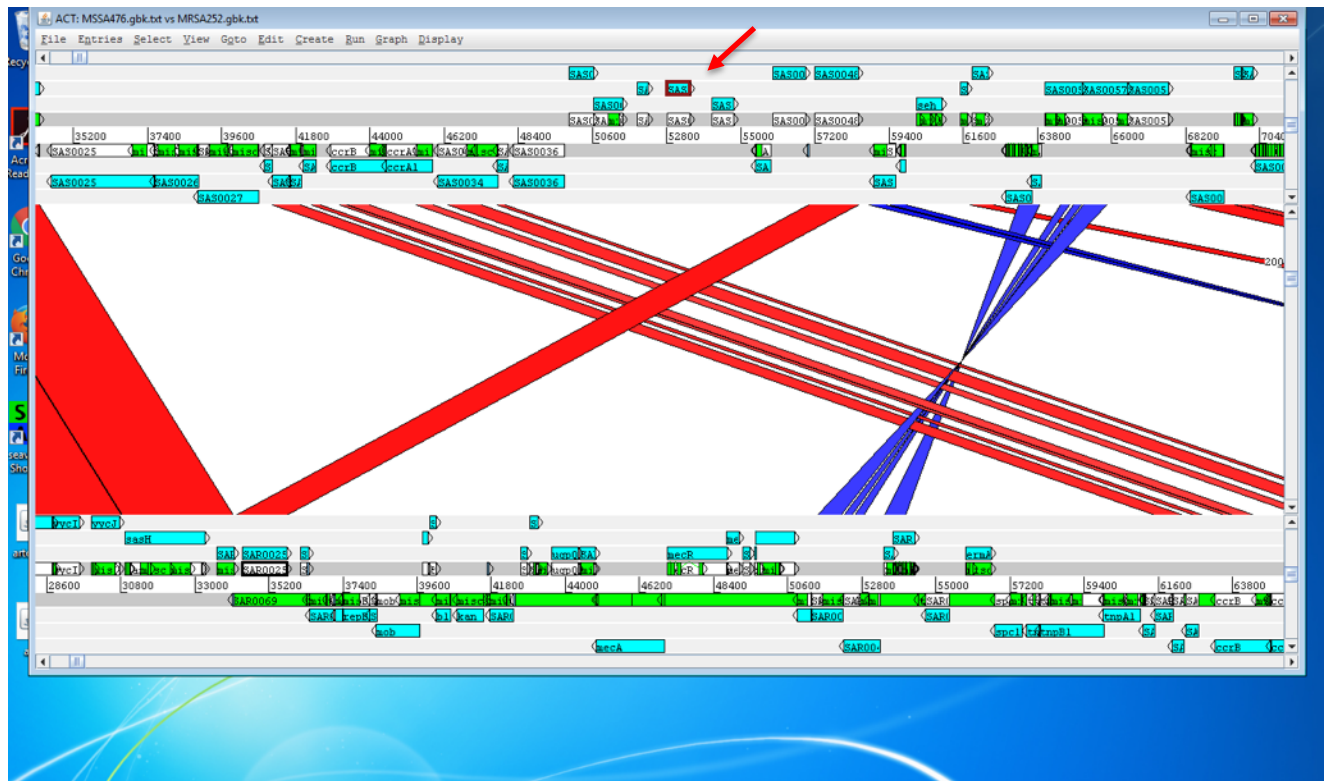
**Step 23:** You can see that there is some degree of synteny between the *SCCmec* element in MRSA252 (bottom) genome and a region in the MSSA476 (top) genome. This is because MSSA476 has another mobile element of the same family called *SCC<sub>476</sub>* (see highlighted region above). Click some of the genes regions of conservation and see what they are (use ctrl-v or right click – View – View selected features). This region is known as the ‘orfX region’ – as SCC elements use a sequence which is part of the *orfX* gene to insert into the genome and is regarded as a hotspot for the acquisition of horizontally transferred DNA.

**Step 24:** You should have found that in both genomes – there are some genes annotated as *ccrA* and *ccrB*. These are the site-specific recombinase genes – that mediated the site-specific integration and excision from the genome, enabling horizontal transfer. Finding these genes alongside resistance genes is common and a useful indicator of how the genes got into the genome.

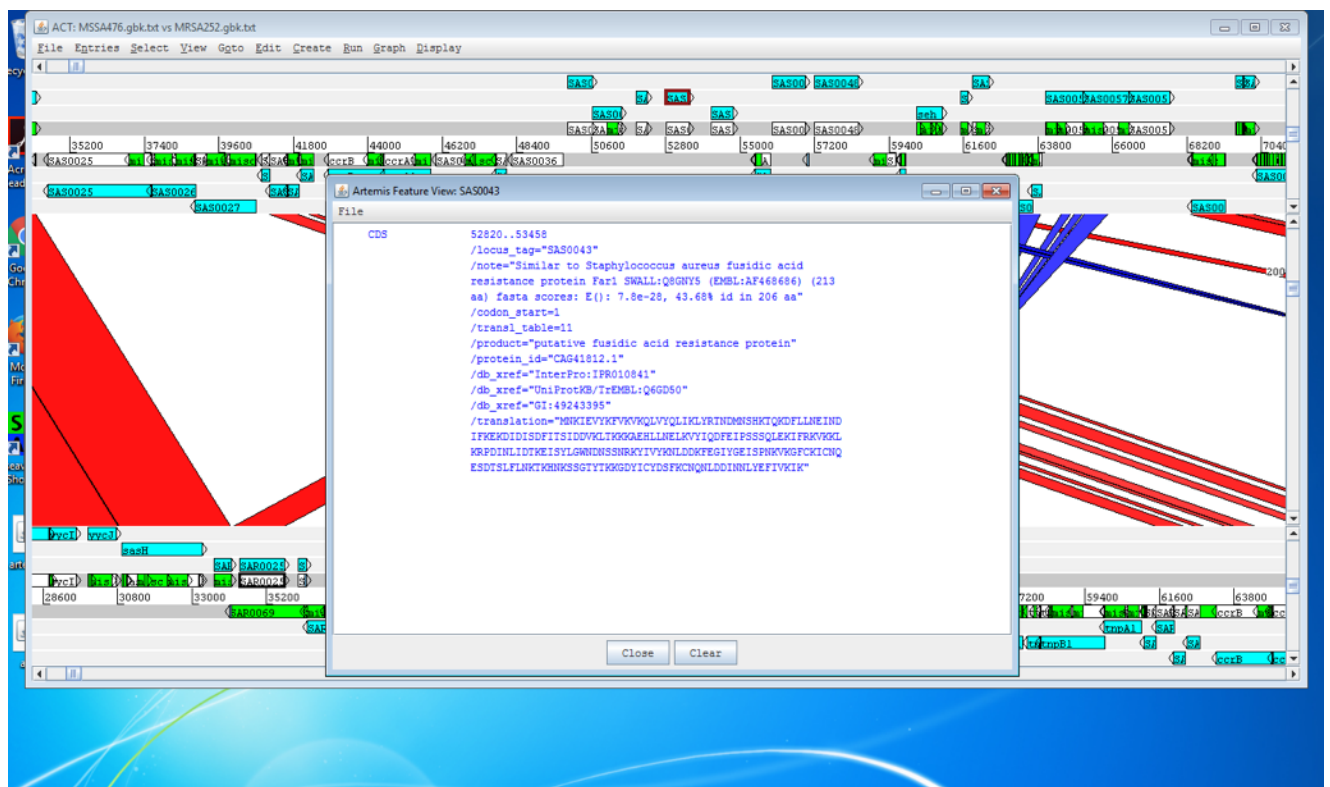




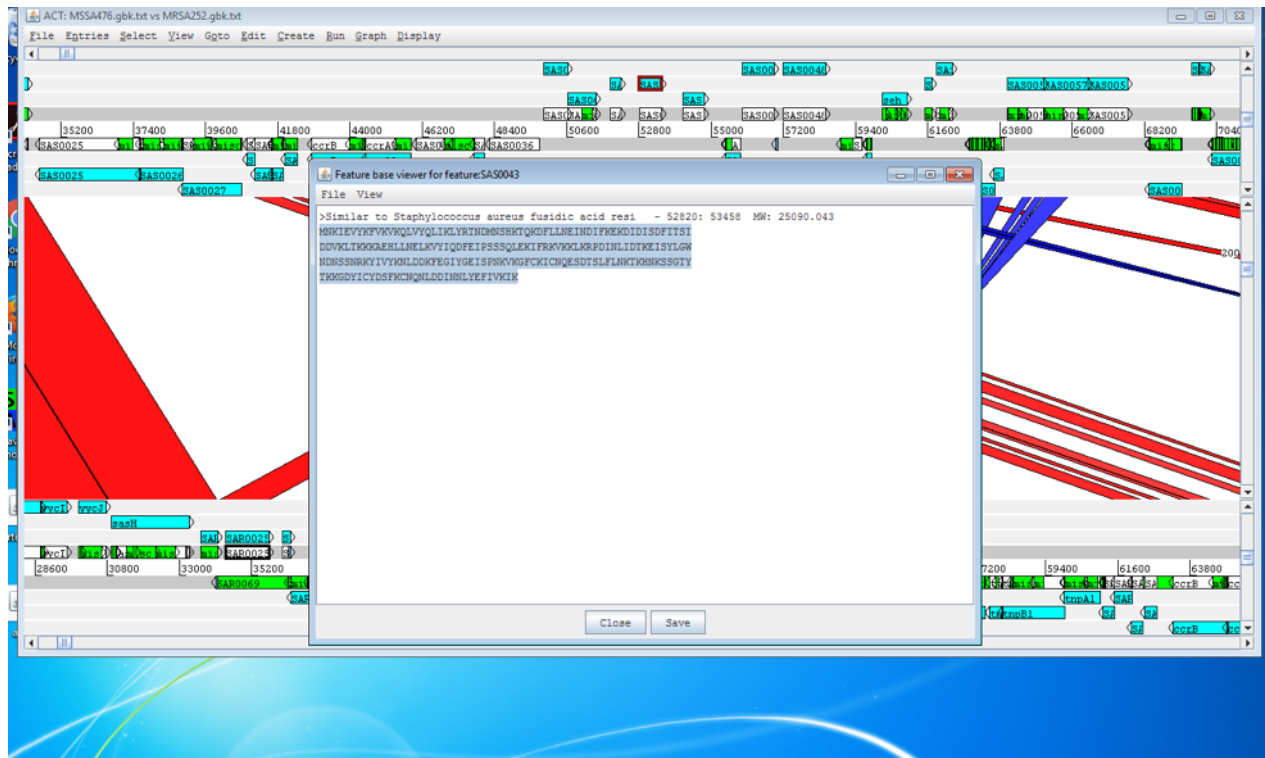




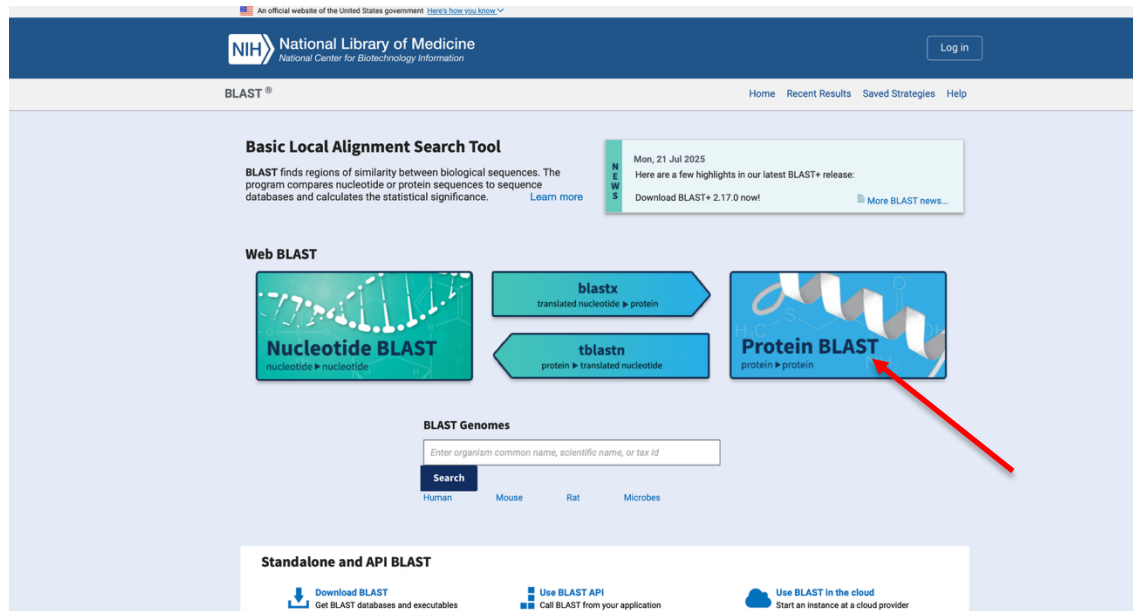
**Step 30:** If you click on the top sequence (MSSA476) and then click on the 'Goto' menu at the top and select the upper sequence 'MSSA476.gbk.txt' and then click 'Navigator'. Click the 'Goto Feature With Gene Name' and type 'SAS0043' in the box and then click the 'Goto' **TWICE** – You should then have a view like above (or something close it). As you can see this gene is not present in the other genome (MRSA252). But as it is in the *orfX* region – it is likely that this gene was also horizontally transferred – but that the element that carried it in has degraded over time.



**Step 31:** Use ctrl-v or right click – ‘View’ – Select the upper sequence ‘MSSA476.gbk.txt’ then ‘View selected features’ – As you can see the gene is annotated as ‘Similar to *Staphylococcus aureus* fusidic acid resistance protein’. We will now check this annotation.



**Step 32:** Now Click on the ‘View’ window and select upper sequence ‘MSSA476.gbk.txt’ ‘Amino Acids’ and then select ‘Amino Acids of Selection As Fasta’ and then highlight and copy the sequence (like shown above) and press ctrl + C to copy the amino acid sequence.



**Step 33:** Now minimise the ACT window and open a web browser. And then type: <https://blast.ncbi.nlm.nih.gov/Blast.cgi> in the web address bar. Then click on ‘Protein blast’ (see red arrow). If you wanted to do this with a nucleotide sequence you would use the Nucleotide BLAST.

**Step 34:** Now paste the sequence into the 'Query sequence' box (see red arrow) and click the BLAST button (blue arrow). Don't worry about any of the other settings will just run things on default.

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Standard Protein BLAST

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Enter Query Sequence

Enter accession number(s), gi(s), or FASTA sequence(s) [Clear](#)

Query subrange

From

To

Or, upload file

Choose File no file selected

Job Title

Enter a descriptive title for your BLAST search

☐ Align two or more sequences

Choose Search Set

Database

ClustalNR (nr\_cluster\_seq)

Organism

Optional

Enter organism name or id—completions will be suggested

Add organism

Enter organism common name, binomial, or tax id. Only 20 top taxa will be shown.

Program Selection

Algorithm

☒ blastp (protein-protein BLAST)

☐ PSI-BLAST (Position-Specific Iterated BLAST)

Choose a BLAST algorithm

BLAST

Search database ClustalNR using Blastp (protein-protein BLAST)

☐ Show results in a new window

+ Algorithm parameters

**Step 35:** You will then get a window like this while the BLAST runs.

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Format Request

Job Title: Protein Sequence

|                       |            |
|-----------------------|------------|
| Request ID            | V32ZYGM016 |
| Status                | Searching  |
| Time since submission | 00:00:00   |

This page will be automatically updated in 2 seconds until search is done

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**Step 36:** Once the BLAST is complete you will get a view like this. The Blue arrow is indicating the section showing that BLAST has found that your amino acid query sequence. The red arrow is showing you the tab you can click on to visualise the alignments and the blast hit scores (the higher the number the better the score and therefore how closely related the BLAST hit score).

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Job Title: Protein Sequence  
RID: V3361MTW014 Search expires on 03-12 23:39 pm [Download All](#)  
Program: BLASTP [Citation](#)  
Database: ClusteredNR [See details](#)  
Query ID: IclQuery\_1153115  
Description: unnamed protein product  
Molecule type: amino acid  
Query Length: 212  
Other reports: [Distance tree of results](#) [Multiple alignment](#) [MSA viewer](#)

**Filter Results**

Organism only top 20 will appear **NEW**  
Type common name, binomial, taxid or group name  
[Add organism](#)

Percent Identity:  to  E value:  to  Query Coverage:  to   
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**Clusters** Graphic Summary **Alignments** Taxonomy

Clusters producing significant alignments Download Select columns Show 100

☒ select all 100 clusters selected [GenPept](#) [Graphics](#) [Distance tree of results](#) [Multiple alignment](#) [MSA Viewer](#)

| Cluster Composition          | Cluster Ancestor      | Cluster Representative Sequence                                        | Max Score | Total Score | Query Cover | E value | Per. Ident | Acc. Len | Accession      |
|------------------------------|-----------------------|------------------------------------------------------------------------|-----------|-------------|-------------|---------|------------|----------|----------------|
| 45 member(s), 11 organism(s) | Staphylococcus gro... | MULTISPECIES: FusC family fusidic acid resistance EF-G-binding p...    | 410       | 410         | 100%        | 2e-144  | 98.11%     | 212      | WP_025176262.1 |
| 6 member(s), 1 organism(s)   | firmicutes            | FusC family fusidic acid resistance EF-G-binding protein [Staphyloc... | 375       | 375         | 89%         | 7e-131  | 100.00%    | 189      | HAY6661289.1   |
| 15 member(s), 2 organism(s)  | firmicutes            | FusC family fusidic acid resistance EF-G-binding protein [Staphyloc... | 362       | 362         | 86%         | 2e-125  | 100.00%    | 183      | HCS9116920.1   |
| 9 member(s), 1 organism(s)   | firmicutes            | FusB/FusC family EF-G-binding protein [Staphylococcus aureus]          | 321       | 321         | 77%         | 1e-109  | 99.39%     | 163      | HCW9121851.1   |
| 4 member(s), 1 organism(s)   | firmicutes            | FusB/FusC family EF-G-binding protein [Macrococcus arment]             | 315       | 315         | 100%        | 2e-106  | 70.75%     | 212      | WP_243365901.1 |
| 7 member(s), 2 organism(s)   | firmicutes            | hypothetical protein [Staphylococcus aureus]                           | 303       | 303         | 73%         | 9e-103  | 100.00%    | 154      | HQC2242244.1   |
| 1 member(s), 1 organism(s)   | firmicutes            | MULTISPECIES: FusB/FusC family EF-G-binding protein [unclassifi...     | 302       | 302         | 100%        | 2e-101  | 66.51%     | 212      | WP_449154410.1 |
| 1 member(s), 1 organism(s)   | firmicutes            | fusaric acid resistance protein, partial [Staphylococcus haemolyticus] | 277       | 277         | 66%         | 2e-92   | 100.00%    | 139      | AIY30227.1     |
| 2 member(s), 1 organism(s)   | firmicutes            | FusB/FusC family EF-G-binding protein [Staphylococcus aureus]          | 271       | 271         | 63%         | 3e-90   | 100.00%    | 134      | HHCH8784653.1  |
| 4 member(s), 4 organism(s)   | firmicutes            | hypothetical protein [Staphylococcus aureus]                           | 263       | 263         | 60%         | 2e-87   | 100.00%    | 123      | MDH1800466.1   |

**Step 37:** If you click on the alignment tab - you will get a view like the one below, which lists all the hits. You can see that most of the top hits label the protein to be involved in Fusidic acid resistance and – and one is labelled '**MULTISPECIES: FusC family fusidic acid resistance EF-G-binding protein [Staphylococcus]**' (See arrow). Have a look at some of the other hits.

Job Title: Protein Sequence  
RID: V3361MTW014 Search expires on 03-12 23:39 pm [Download All](#)  
Program: BLASTP [Citation](#)  
Database: ClusteredNR [See details](#)  
Query ID: IclQuery\_1153115  
Description: unnamed protein product  
Molecule type: amino acid  
Query Length: 212  
Other reports: [Distance tree of results](#) [Multiple alignment](#) [MSA viewer](#)

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Organism only top 20 will appear **NEW**  
Type common name, binomial, taxid or group name  
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**Alignments** Taxonomy

Alignment view: Pairwise [Restore defaults](#) Download

100 clusters selected

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**MULTISPECIES: FusC family fusidic acid resistance EF-G-binding protein [Staphylococcus]**  
Sequence ID: WP\_025176262.1 Length: 212 Number of Matches: 1

Range 1: 1 to 212 [GenPept](#) [Graphics](#) [Next Match](#) [Previous Match](#)

| Score          | Expect                                                    | Method                       | Identities   | Positives    | Gaps      |
|----------------|-----------------------------------------------------------|------------------------------|--------------|--------------|-----------|
| 410 bits(1055) | 2e-144                                                    | Compositional matrix adjust. | 209/212(98%) | 209/212(98%) | 0/212(0%) |
| Query 1        | MNKEIVYKFKVKQVLYQILKLYRTNDNNSHKTDQKFLNEINDIFKEKDIDSDFTTSI | 60                           |              |              |           |
| Sbjct 1        | MNKEIVYKFKVKQVLYQILKLYRTNDNNSHKTDQKFLNEINDIFKEKDIDSDFTTSI | 60                           |              |              |           |
| Query 61       | DDVKLTKKKAHLLNELKVIYIDFEPSSQLEKIFRKVKKLRPDINDITKEISYLGW   | 120                          |              |              |           |
| Sbjct 61       | DDVKLTKKKAHLLNELKVIYIDFEPSSQLEKIFRKVKKLRPDINDITKEISYLGW   | 120                          |              |              |           |
| Query 121      | NDNSNRKYIYVKNLDDKFEIYGEISPNKVGFKCIQNESDTSFLNKTQHNKSSGT    | 180                          |              |              |           |
| Sbjct 121      | NDNSNRKYIYVKNLDDKFEIYGEISPNKVGFKCIQNESDTSFLNKTQHNKSSGT    | 180                          |              |              |           |
| Query 181      | TKKGDYICYSFKCNLDDNNLYEFIVTK                               | 212                          |              |              |           |
| Sbjct 181      | TKKGDYICYSFKCNLDDNNLYEFIVTK                               | 212                          |              |              |           |

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Program: BLASTP [Citation](#)

Database: pdb [See details](#)

Query ID: lcl|Query\_10600988

Description: Similar to Staphylococcus aureus fusidic acid resi - 52820 ...

Molecule type: amino acid

Query Length: 212

Other reports: [Distance tree of results](#) [Multiple alignment](#) [MSA viewer](#)

Organism: only top 20 will appear ☐ exclude

Type common name, binomial, taxid or group name

[+ Add organism](#)

Percent Identity:  to  E value:  to  Query Coverage:  to

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Alignment view: Pairwise [Restore defaults](#) [Download](#)

5 sequences selected

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**Chain A, PUTATIVE FUSIDIC ACID RESISTANCE PROTEIN [Staphylococcus aureus]**

Sequence ID: **2YB5\_A** Length: 215 Number of Matches: 1

[See 1 more title\(s\)](#) [See all Identical Proteins \(IPG\)](#)

Range 1: 4 to 215 [GenPept](#) [Graphics](#) [Next Match](#) [Previous Match](#)

| Score          | Expect                                                       | Method                       | Identities    | Positives     | Gaps      |
|----------------|--------------------------------------------------------------|------------------------------|---------------|---------------|-----------|
| 419 bits(1077) | 5e-151                                                       | Compositional matrix adjust. | 212/212(100%) | 212/212(100%) | 0/212(0%) |
| Query 1        | MNKIEVYKFKVKQLVYQLIKLYRTNDMNSHKTQKDFLLNEINDIFKEKDIDISDFITSI  |                              | 60            |               |           |
| Sbjct 4        | MNKIEVYKFKVKQLVYQLIKLYRTNDMNSHKTQKDFLLNEINDIFKEKDIDISDFITSI  |                              | 63            |               |           |
| Query 61       | DDVKLTKKKAEHLLNELKVIYIDFEIPSSSQLEKIFRKVKKLKRPDINLIDTKEISYLGW |                              | 120           |               |           |
| Sbjct 64       | DDVKLTKKKAEHLLNELKVIYIDFEIPSSSQLEKIFRKVKKLKRPDINLIDTKEISYLGW |                              | 123           |               |           |
| Query 121      | NDNNSNRKYIVYKNLDDKFEIGYGEISPNKVGKFCIKNQESDTSFLNKTTHNKSSGTY   |                              | 180           |               |           |
| Sbjct 124      | NDNNSNRKYIVYKNLDDKFEIGYGEISPNKVGKFCIKNQESDTSFLNKTTHNKSSGTY   |                              | 183           |               |           |
| Query 181      | TKKGDYICYDSFKCNQLDDINNLYEFIVKIK                              |                              | 212           |               |           |
| Sbjct 184      | TKKGDYICYDSFKCNQLDDINNLYEFIVKIK                              |                              | 215           |               |           |

**Related Information**

[Structure](#) - 3D structure displays

[Identical Proteins](#) - Identical proteins to 2YB5\_A

This shows you that the sequence that we BLASTed is 100% identical to this entry in the database. If you click the link indicated by the red arrow it will take you to the protein database entry for this hit.

NCBI Blast: Protein Sequence - Chain A, Structure Of The Fusidic Acid Resistance Protein Fusc

Protein: Protein [Search](#)

[GenPept](#) [FASTA](#) [Graphics](#)

**Chain A, Structure Of The Fusidic Acid Resistance Protein Fusc**

PDB: 2YB5\_A

[Identical Proteins](#) [FASTA](#) [Graphics](#)

[Go to:](#)

LOCUS: 2YB5\_A, 215 aa, linear, BCT 23-SEP-2015

DEFINITION: Chain A, Putative Fusidic Acid Resistance Protein.

ACCESSION: 2YB5\_A

VERSION: 2YB5\_A

DESOURCE: pdb: molecule 2YB5, chain 65, release Sep 23, 2015; deposition: Mar 1, 2011; class: Translation; source: Mmdb\_id: 96621, Pdb\_id 1: 2YB5; Exp. method: X-Ray Diffraction.

KEYWORDS: Staphylococcus aureus

SOURCE ORGANISM: [Staphylococcus aureus](#)

REFERENCE: 1 (residues 1 to 215)

AUTHORS: Cox, G., Thompson, G.S., Jenkins, H.T., Peske, F., Savelsbergh, A., Rodina, M.V., Wintermeyer, W., Homans, S.W., Edwards, T.A. and O'Neill, A.J.

TITLE: Ribosome clearance by FusB-type proteins mediates resistance to the antibiotic fusidic acid

JOURNAL: Proc. Natl. Acad. Sci. U.S.A. 109 (6), 2102-2107 (2012)

PUBMED: 22388410

Send to: [Change region shown](#) [Customize view](#)

**Analyze this sequence**

[Run BLAST](#)

[Identify Conserved Domains](#)

[Highlight Sequence Features](#)

[Find in this Sequence](#)

**Protein 3D Structure**

 Structure Of The Fusidic Acid Resistance Protein Fusc

PDB: 2YB5

Source: Staphylococcus aureus

Method: X-Ray Diffraction

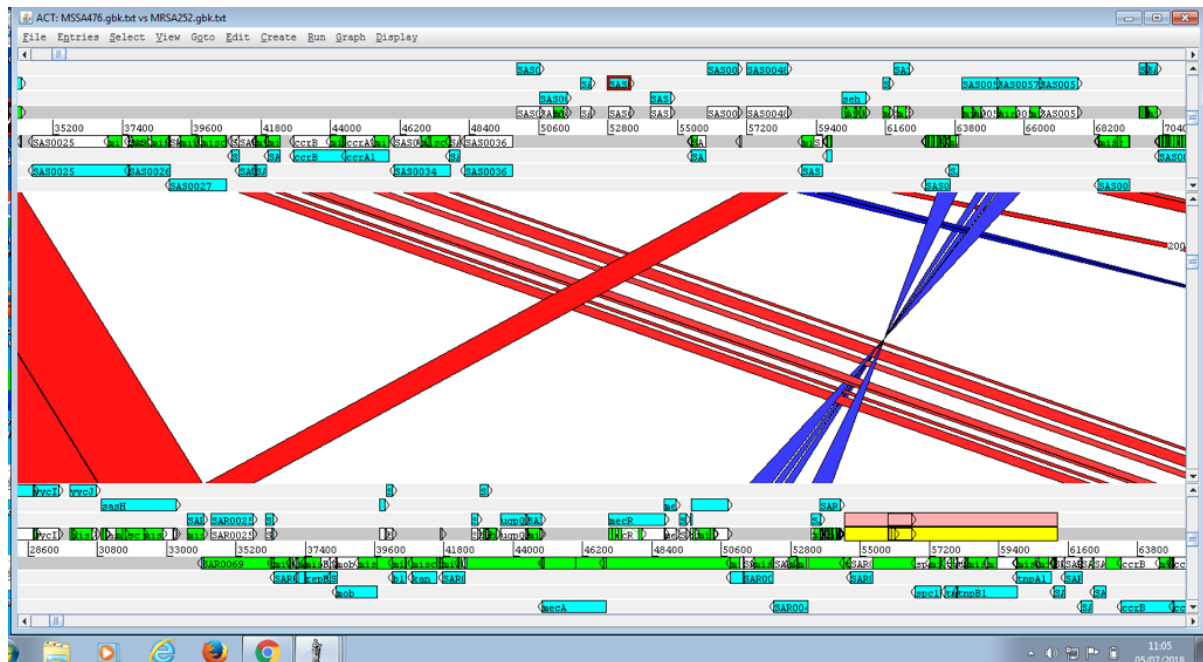
Resolution: 2.1 Å

**Related information**

[Similar protein sequences using SmartBlast](#)

**Step 39:** This entry shows that there is experimental evidence that this protein is involved in resistance to Fusidic acid – so we can be quite confident that this protein is involved in resistance to Fusidic acid.

If you now go back and have a look at some of the other hits on your first blastp, you will see that the information available – can be considerable vaguer – with proteins annotated as ‘hypothetical protein’ or ‘unnamed protein product’.



**Step 40:** Now return to the ACT view. The SCCmec in MRSA252 contains some more resistance genes – that are highlighted above – using what you have just learned look up the annotation of these genes and check the annotations using BLAST. Which resistance genes are present in this region? What other genes are present? Do you have any idea of how these genes might have got in to the SCCmec element?



## Part II: Investigating multidrug resistance plasmids from the CPE outbreak

In the previous two parts of this practical you have been looking at completed genomes – that is genomes that have been completely sequenced, aligned into a single chromosome and manually annotated (that is each gene annotation checked manually). But if you generate your own data, you will probably end up with a less ‘polished’ genome to work with.

In this practical you are going to look at two of the plasmids from two of the outbreak (these are in the folder Practical 3):

1. CP004 – *Klebsiella pneumoniae*
2. CP0069 – *E. coli*

Both the plasmids from these strains have the same *bla*<sub>NDM-1</sub> gene, we are going to use ACT to compare the plasmid sequences to see how much homology the two plasmids share and to determine any structural features that might be of importance.

**Step 1:** Firstly, we need to generate the BLAST comparison file that ACT takes to show how related the two sequences you are comparing. Open: <https://blast.ncbi.nlm.nih.gov/Blast.cgi> in a web browser (Firefox).

You should get a window like this:

The screenshot shows the NCBI BLAST website. At the top is the NIH National Library of Medicine header with a 'Log in' button. Below the header is a navigation bar with 'BLAST®' and links for 'Home', 'Recent Results', 'Saved Strategies', and 'Help'. The main content area is titled 'Basic Local Alignment Search Tool' and includes a brief description of BLAST. To the right of this is a 'NEWS' box with a message about non-interactive searches. Below the main title is the 'Web BLAST' section, which features three large buttons: 'Nucleotide BLAST' (nucleotide to nucleotide), 'blastx' (translated nucleotide to protein), and 'tblastn' (protein to translated nucleotide). To the right of these is a 'Protein BLAST' button (protein to protein). At the bottom is the 'BLAST Genomes' section, which has a search input field and a 'Search' button, with links for 'Human', 'Mouse', 'Rat', and 'Microbes' below it.

**Step 2:** Now click on 'Nucleotide BLAST'

You should get a window like this

The screenshot shows the BLAST suite interface. At the top, there's a header 'BLAST® » blastn suite' and a sub-header 'Align Sequences Nucleotide BLAST'. Below this, there are tabs for 'blastn', 'blastp', 'blastx', 'tblastn', and 'tblastx'. The 'blastn' tab is selected. The main area is divided into three sections: 'Enter Query Sequence', 'Enter Subject Sequence', and 'Program Selection'. In the 'Enter Query Sequence' section, there's a text input field for 'Enter accession number(s), gi(s), or FASTA sequence(s)', a 'Clear' button, and a 'Query subrange' section with 'From' and 'To' input fields. Below this, there's a section for 'Or, upload file' with a 'Choose File' button and 'no file selected' text. A 'Job Title' input field is also present. In the 'Enter Subject Sequence' section, there's a similar text input field, 'Clear' button, 'Subject subrange' section, and 'Or, upload file' section. The 'Program Selection' section has 'Optimize for' radio buttons: 'Highly similar sequences (megablast)' (selected), 'More dissimilar sequences (discontiguous megablast)', and 'Somewhat similar sequences (blastn)'. A 'Choose a BLAST algorithm' link is also there. A red arrow points to the 'Align two or more sequences' checkbox, which is checked.

**Step 3:** Click the tick box where it says 'Align two or more sequences' (see red arrow above). This will allow a comparison of two sequences.

This screenshot shows the same BLAST suite interface as before, but with files selected for both query and subject sequences. In the 'Enter Query Sequence' section, the 'Choose File' button is highlighted with a green arrow, and the file 'cpe001\_pla...l1B.fasta' is displayed. In the 'Enter Subject Sequence' section, the 'Choose File' button is highlighted with a blue arrow, and the file 'cpe069\_plasmid\_IncAC.fasta' is displayed. The 'Align two or more sequences' checkbox remains checked. The 'Program Selection' section is unchanged. At the bottom, there's a 'BLAST' button and a checkbox for 'Show results in a new window'.

**Step 4:** Now click the 'choose file' on the upper section (green arrow) and select the file – 'cpe004\_plasmid\_IncFIB-HI1B.fasta' (you will need to navigate to the file location). Do the same for the lower sequence (blue arrow) and select – 'cpe069\_plasmid\_IncAC.fasta'. Then press BLAST (blue button)

**Step 5:** You should now get a view like this. You need to click on the alignment tab (see the red arrow below)

National Library of Medicine  
National Center for Biotechnology Information

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BLAST® » blastn suite-2sequences » results for RID-U65S56RT114

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Job Title: **tig00000008\_np1212 358845**  
RID: [U65S56RT114](#) Search expires on 02-06 14:45 pm [Download All](#)  
Program: Blast 2 sequences [Citation](#)  
Query ID: lcl|Query\_2194685 (dna)  
Query Descr: tig00000008\_np1212 358845  
Query Length: 358845  
Subject ID: lcl|Query\_2194687 (dna)  
Subject Descr: cpe069\_hybrid\_plasmid-incAC Escherichia coli strain strain  
Subject Length: 145843  
Other reports: [MSA viewer](#)

**Filter Results**  
Percent Identity:  to   
E value:  to   
Query Coverage:  to   
[Filter](#) [Reset](#)

Descriptions
Graphic Summary
Alignments
Dot Plot

Sequences producing significant alignments
Download Select columns Show 100

☒ select all 1 sequences selected

[Graphics](#)
[MSA Viewer](#)

|                                     | Description                                                                | Scientific Name | Max Score | Total Score | Query Cover | E value | Per. Ident | Acc. Len | Accession     |
|-------------------------------------|----------------------------------------------------------------------------|-----------------|-----------|-------------|-------------|---------|------------|----------|---------------|
| <input checked="" type="checkbox"/> | <a href="#">cpe069_hybrid_plasmid-incAC Escherichia coli strain strain</a> |                 | 11705     | 1.093e+05   | 17%         | 0.0     | 99.97%     | 145843   | Query_2194687 |

**Step 6:** You should now see a view like this – select ‘**Download**’ from the righthand side (red arrow) and select ‘Hit Table (text)’. This will generate and download a file that you can read into ACT. Note – where this file has downloaded as you are going to need it. It will be called ‘xxxxxxxx--Alignment.txt’. (if you have problems there is a folder called (backup – which contains one we have pre-generated).

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BLAST® » blastn suite-2sequences » results for RID-U65S56RT114

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Job Title: **tig00000008\_np1212 358845**  
RID: [U65S56RT114](#) Search expires on 02-06 14:45 pm [Download All](#)  
Program: Blast 2 sequences [Citation](#)  
Query ID: lcl|Query\_2194685 (dna)  
Query Descr: tig00000008\_np1212 358845  
Query Length: 358845  
Subject ID: lcl|Query\_2194687 (dna)  
Subject Descr: cpe069\_hybrid\_plasmid-incAC Escherichia coli strain strain  
Subject Length: 145843  
Other reports: [MSA viewer](#)

**Filter Results**  
Percent Identity:  to   
E value:  to   
Query Coverage:  to   
[Filter](#) [Reset](#)

Descriptions
Graphic Summary
Alignments
Dot Plot

Alignment view: Pairwise ☐ CDS feature [Restore defaults](#)

Download

1 sequences selected

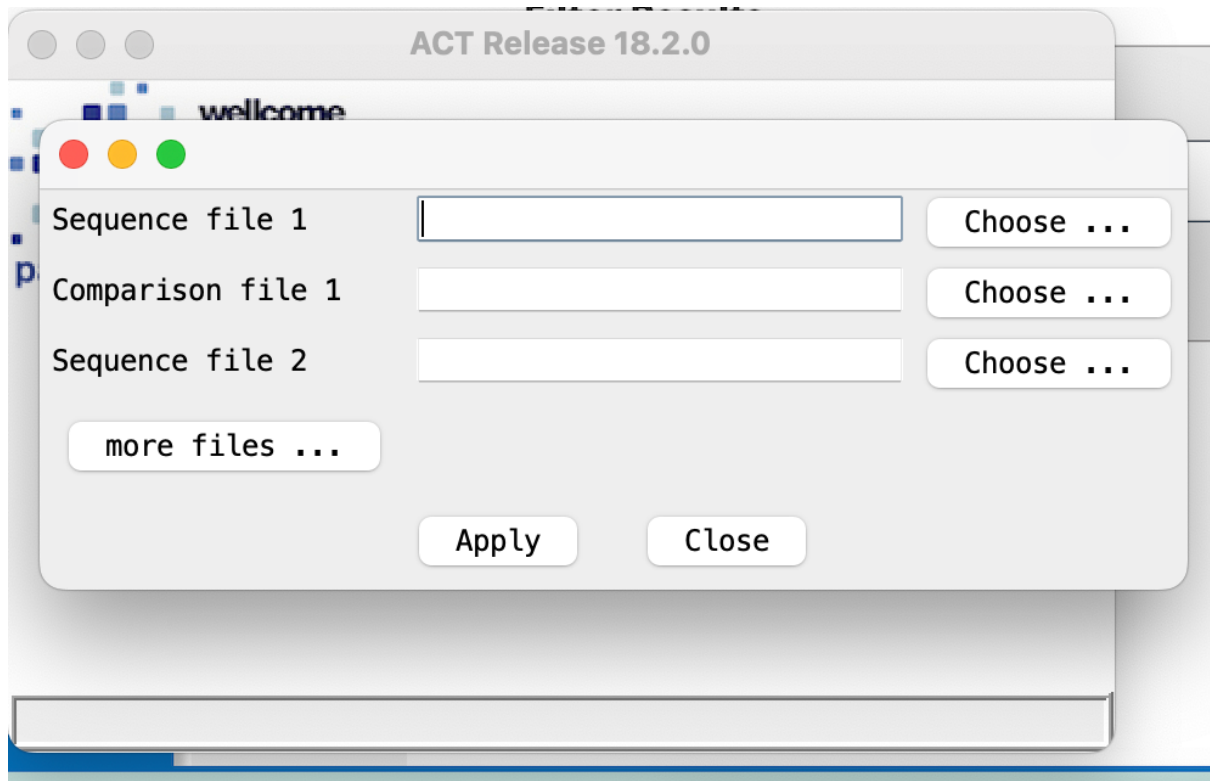
[Download](#)
[Graphics](#)
Sort by: E value

Next
Previous

**cpe069\_hybrid\_plasmid-incAC Escherichia coli strain strain**  
Sequence ID: [Query\\_2194687](#) Length: 145843 Number of Matches: 40  
Range 1: 115022 to 121365 [Graphics](#)

Next Match
Previous Match

FASTA (completesequences)  
FASTA (aligned sequences)  
**Hit Table (text)**  
Hit Table (CSV)  
Text  
XML  
ASN.1

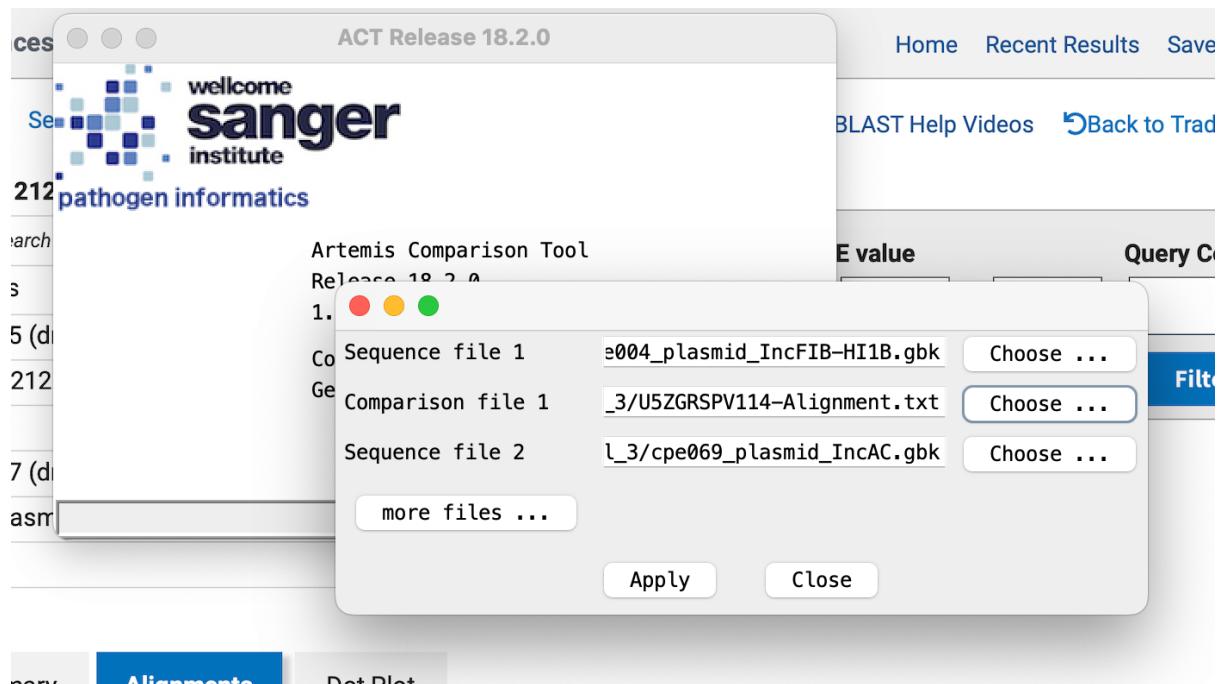


**Step 7:** Open ACT and load the following into:

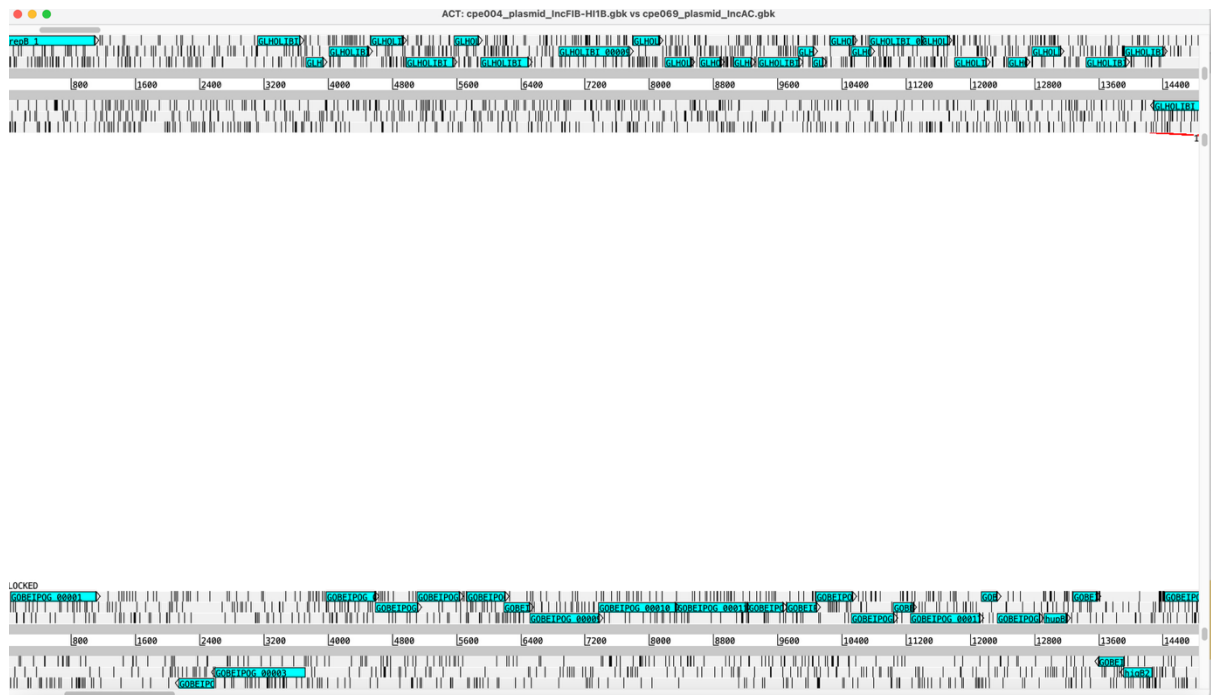
- (a) Sequence file 1: 'cpe004\_plasmid\_IncFIB-HI1B.gbk'
- (b) Comparison file 1: 'xxxxxxx--Alignment.txt' (this is file you just downloaded).
- (c) Sequence file 2: 'cpe069\_plasmid\_IncAC.gbk'

Note – we are now using the .gbk files as they contain the annotation information.

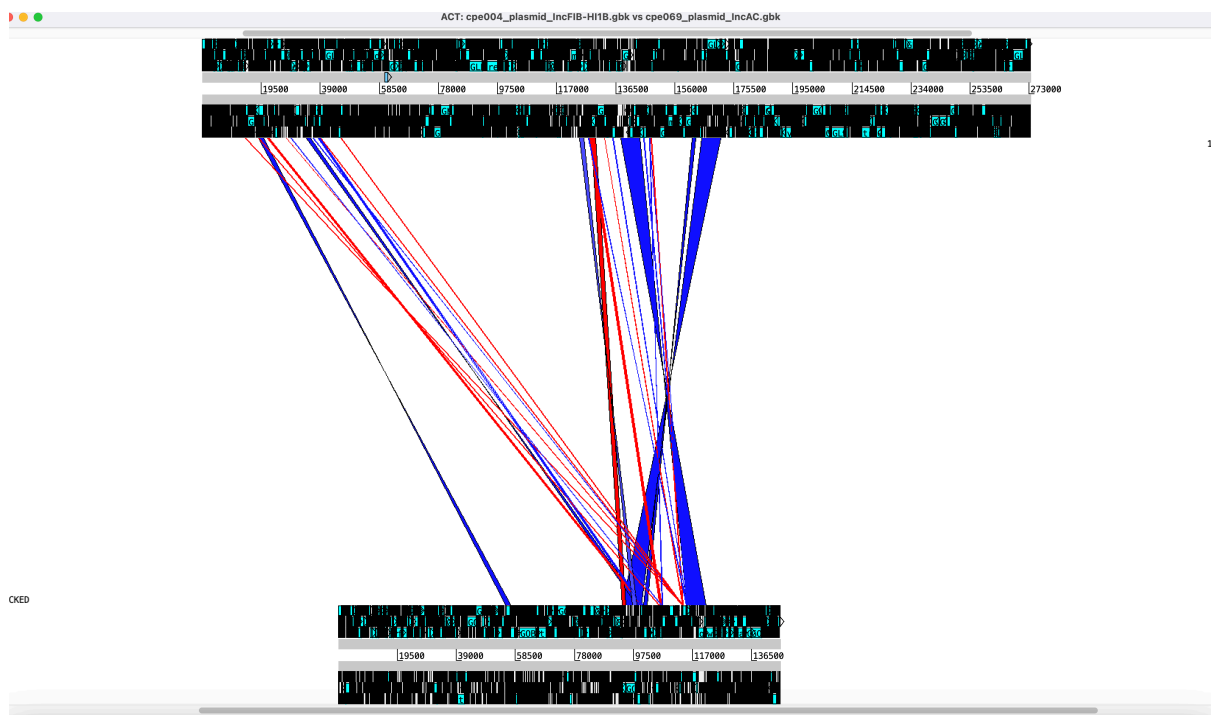
The image below shows what it should look like.



**Step 8:** You should now have a view like this. Remember to unlock the two sequences and then zoom out and take a look at the two sequence.

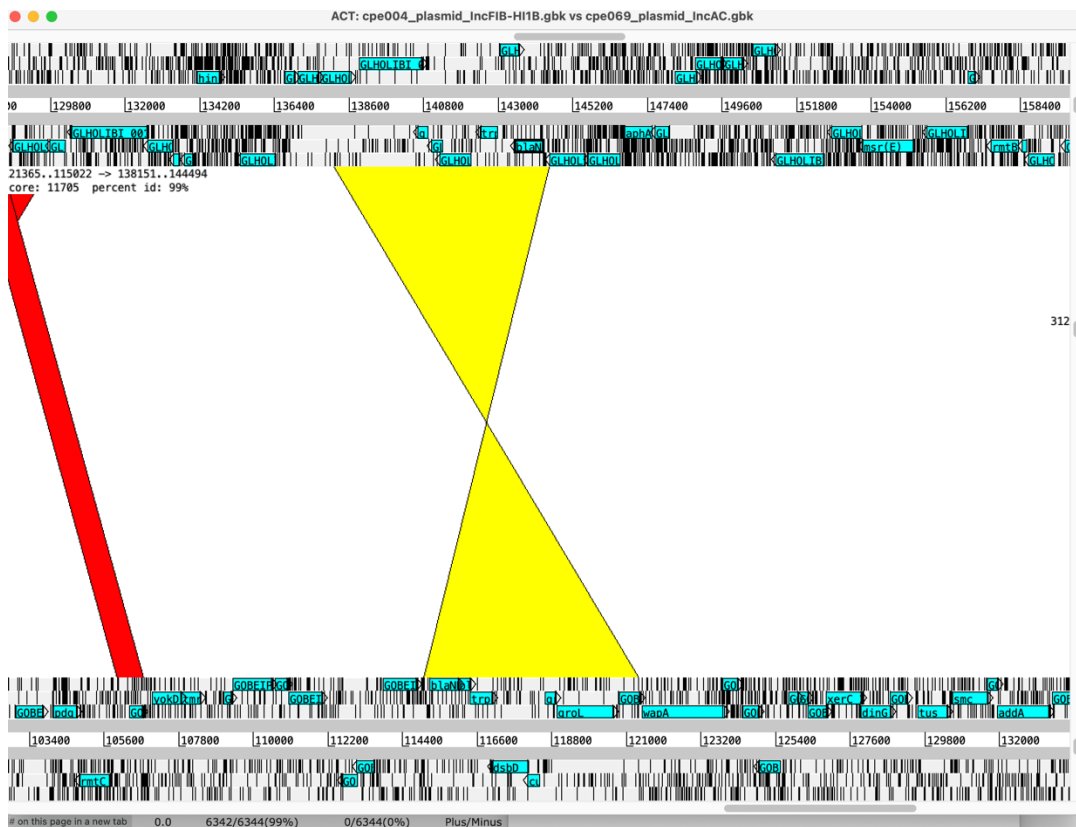
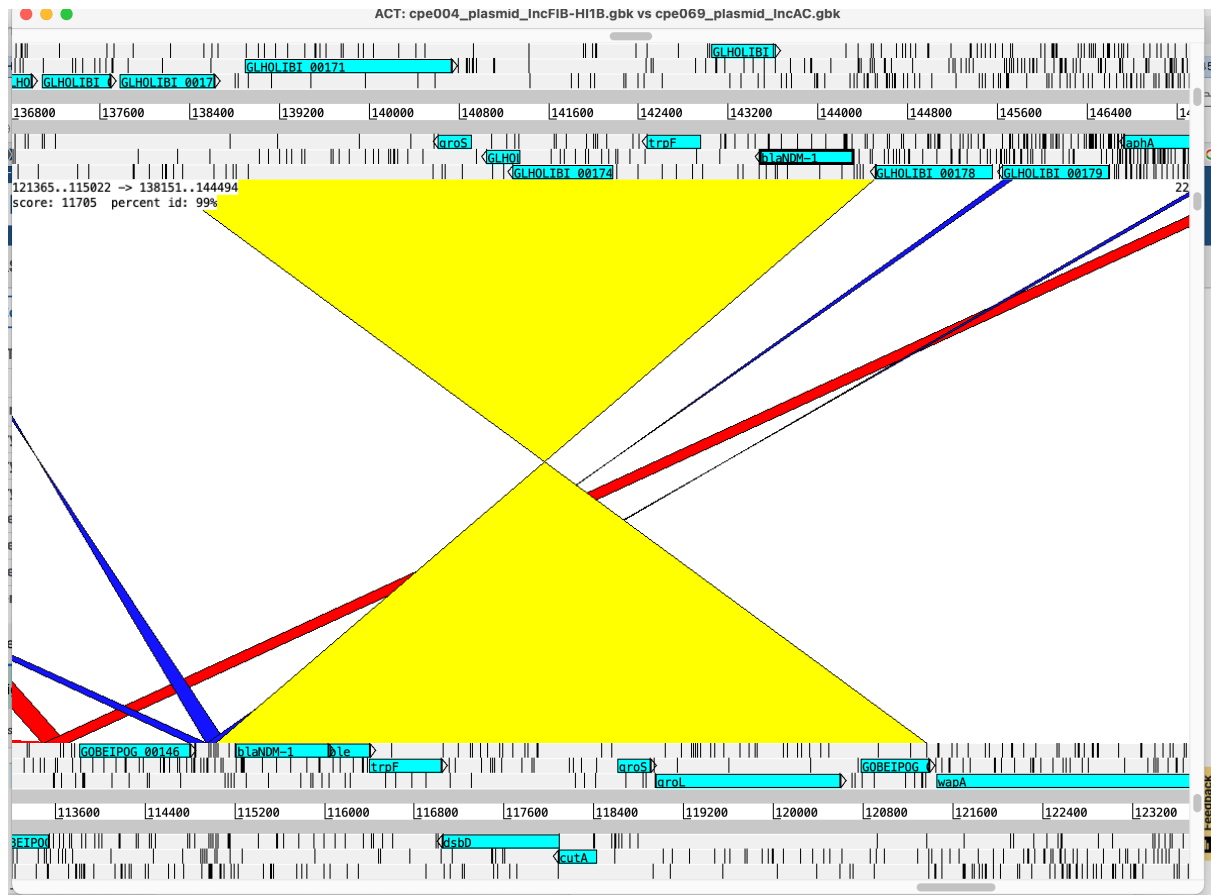


**Step 8:** You should have a view like this. You can clearly see although the two plasmids are generally distinct, they do share some blocks of homologous sequences.

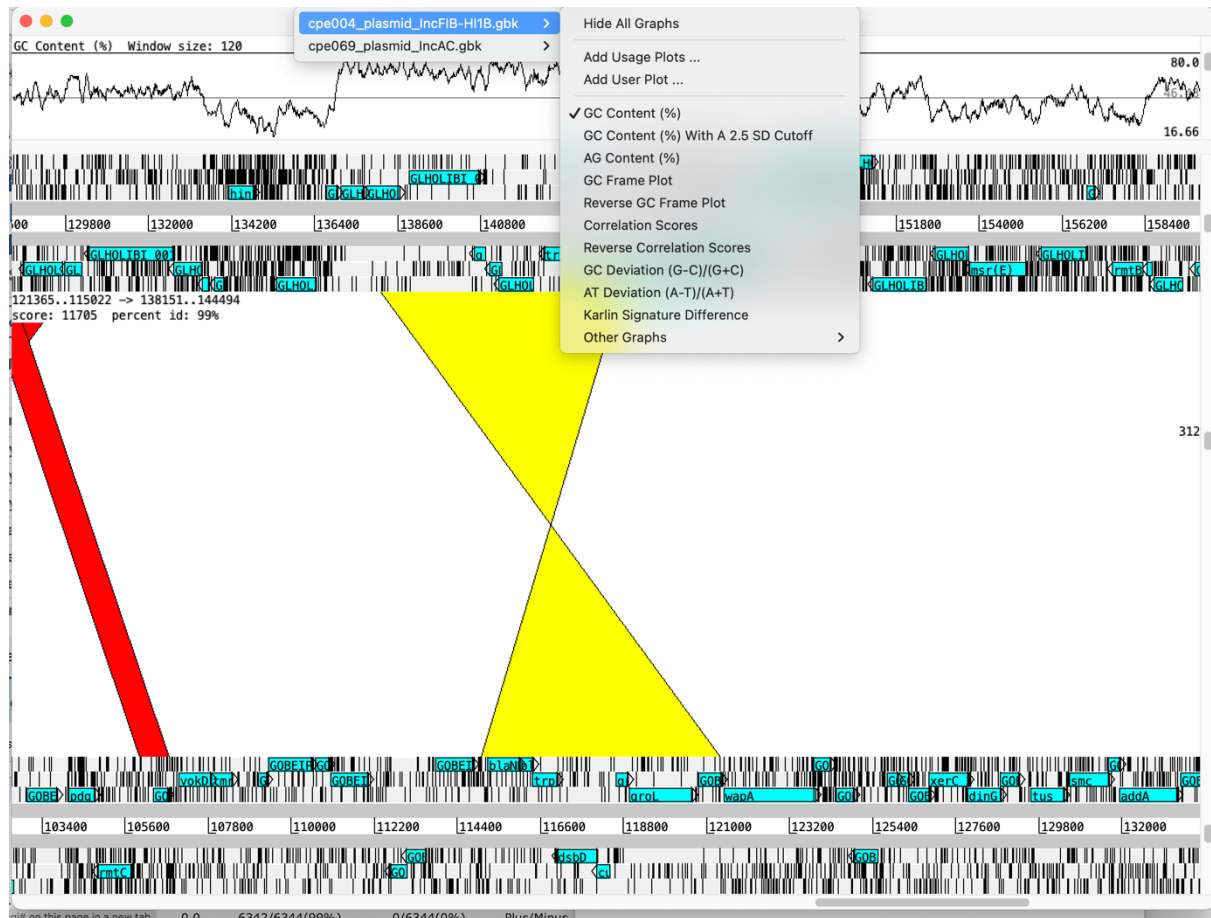


**Step 9:** If you use the navigator (control/command – G) or on the menu ‘Goto’ and select ‘cpe004\_plasmid\_IncFIB-HI1B.gbk’ – Navigator to use the ‘Goto Feature with Gene Name’ and type: ‘ndm’ (to take you to the *bla*<sub>NDM-1</sub> gene. Then zoom in to this area (like shown below). You can clearly see the two plasmids both share the same block.





**Step 10:** What genes are upstream of the *bla*<sub>NDM-1</sub> in both plasmids? How might they have contributed to the presence of *bla*<sub>NDM-1</sub> ?



**Step 11:** Go to the 'Graph' menu at the top for the cpe004\_plasmid\_incFIB-HI1B.gb' and select 'GC content (%)'. You should then see a graph appear at the top.

